Multi-omics Integration of the TCGA-LIHC

Transcriptomic, Epigenomic, and Post-transcriptomic Profiling via MOFA

Course: Integrative Bioinformatics Group-E

ABSTRACT

Hepatocellular carcinoma (HCC) is among the most prevalent and lethal malignancies worldwide, driven by a molecularly heterogeneous pathogenesis that cannot be adequately captured by any single data modality. Here we present a structured multi-omics integration study of the TCGA Liver Hepatocellular Carcinoma (TCGA-LIHC) cohort, profiling three complementary molecular layers — bulk RNA-seq transcriptomics, Illumina HumanMethylation450K DNA methylation, and microRNA expression — across up to 424 patient samples, with differential analyses contrasting DECEASED versus LIVING patients throughout. RNA-seq analysis via *limma* on 20,531 genes identified two significantly upregulated transcripts in deceased patients: *EPO* ($\log_2$FC = +1.29), a pro-tumorigenic cytokine promoting apoptosis resistance, and *S100A9* ($\log_2$FC = +0.85), a damage-associated molecular pattern protein mediating tumor microenvironment remodeling. DNA methylation profiling of 319,133 high-confidence CpG probes across 377 tumour samples identified 401 significant differentially methylated positions (39 hypermethylated, 362 hypomethylated in deceased patients). miRNA profiling via DESeq2 across 424 samples yielded 46 significantly dysregulated microRNAs, including *hsa-miR-3170*, *hsa-miR-424*, and *hsa-miR-326* as top upregulated candidates; functional enrichment identified a coordinated 22-miRNA apoptosis-evasion hub, hepatocyte-specific regulatory modules, and pharmacogenomic intersections with Gemcitabine and 5-FU. Integration via Multi-Omics Factor Analysis (MOFA) .

Keywords: Hepatocellular carcinoma · TCGA-LIHC · RNA-seq · DNA methylation · microRNA · MOFA · Differential expression

1. Introduction

Hepatocellular carcinoma (HCC) is the most common primary liver malignancy and the third leading cause of cancer-related mortality globally. To understand the HCC heterogeneity among patients; The Cancer Genome Atlas Liver Hepatocellular Carcinoma project (TCGA-LIHC) provides one of the most comprehensively annotated multi-platform molecular datasets for HCC, comprising matched transcriptomic, epigenomic, genomic, and clinical data across hundreds of patients. The cohort clinical and phenotypic metadata were sourced from the TCGA-LIHC phenotype file (`liver.txt`), encompasses 438 sample records with variables including vital status, sex, pathological stage (I–IV), age at diagnosis, and overall survival data. The primary clinical contrast of interest throughout all omics analyses is **vital status**: DECEASED versus LIVING, used as a binary surrogate for prognosis.

A key challenge in HCC research is that individual omics layers capture only a partial view of tumor biology. Gene expression patterns are shaped by upstream epigenetic regulation, including promoter DNA methylation and post-transcriptional silencing by microRNAs. Conversely, miRNA dysregulation can reprogram transcriptional landscapes without direct changes in the genomic sequence. Achieving a mechanistic understanding of survival-associated molecular alterations therefore requires the integration of these complementary data modalities.

In this report, we present a structured multi-omics analysis of the TCGA-LIHC cohort. We first characterize each molecular layer independently through differential expression analyses and functional enrichment. We then apply Multi-Omics Factor Analysis (MOFA), a principled statistical framework that learns latent factors capturing co-variation across omics layers, to integrate RNA-seq, DNA methylation, and miRNA data into a unified low-dimensional representation. Together, these analyses aim to identify molecular signatures associated with patient survival and to reveal the cross-modal regulatory architecture underlying HCC aggressiveness.

2. Data and Experimental Setup

All molecular and clinical data were obtained from the TCGA-LIHC project and accessed via the UCSC Xena browser and the GDC data portal. The dataset encompasses three omics modalities profiled on overlapping sets of patient samples:

Table 1. TCGA-LIHC omics data overview.

Omics Layer	Data Type	Features	Samples	Format
RNA-seq	Gene expression (FPKM)	20,531	423	Pre-normalized continuous
DNA Methylation	β-values (Illumina 450K)	319,133	377	Beta values (0–1)

miRNA-seq	miRNA expression (counts)	1,046	424	Raw integer counts

3. Materials and Methods

3.1 Analytical Framework Overview

All analyses were conducted in R (version 4.5.1). The analytical pipeline follows a modular structure: each omics layer is processed independently through preprocessing, quality assessment, dimensionality reduction, and differential analysis before being passed to the integrative MOFA layer. The general workflow proceeds as follows:

1. Raw data ingestion and sample harmonization across modalities
2. Variance-stabilizing transformation and quality control
3. Dimensionality reduction via PCA for exploratory visualization
4. Differential analysis comparing DECEASED vs. LIVING patients within each modality
5. Functional enrichment and pathway analysis of significant features
6. Multi-omics factor analysis (MOFA) integrating all three layers

3.2 R Packages

The following R packages were employed across the analysis pipeline:

Package	Purpose
`limma`	Linear models for differential expression (RNA-seq, methylation)
`DESeq2`	Count-based DE analysis (miRNA)
`ggplot2`	Data visualization
`ggrepel`	Non-overlapping volcano plot labels
`pheatmap`	Hierarchical clustering heatmaps
`gprofiler2`	GO / KEGG / Reactome enrichment (RNA-seq)
`RColorBrewer`	Colour palette specification
`MOFA2`	Multi-omics factor analysis
`missMethyl`	Bias-corrected GO/KEGG enrichment for methylation
`minfi`	Methylation array processing
`readxl`	Excel metadata ingestion
`data.table`	Fast data loading for large methylation matrices

3.3 RNA-seq Analysis

3.3.1 Dataset and Preprocessing

Gene expression data were obtained from the TCGA-LIHC project. The expression matrix (exp file) comprised 20,531 gene features measured across 423 samples provided as pre-normalized FPKM (Fragments Per Kilobase per Million mapped reads) values. This normalization corrects two sources of bias inherent to RNA-seq data: Sequencing depth and gene length biases.

Sample barcodes in the expression matrix (dot-delimited, e.g., TCGA.2V.A95S.01) were converted to standard TCGA hyphen-delimited format (TCGA-2V-A95S-01) using regular expression substitution in `gsub()` in R. FPKM values were $\log_2$-transformed as $\log_2(\text{FPKM} + 1)$ to stabilize variance and approximate a normal distribution suitable for linear modelling.

3.3.2 Quality Control

Quality control was performed at multiple levels:

- Distribution assessment: Histograms of mean log2(FPKM+1) expression were plotted to visualize the bimodal distribution characteristic of RNA-seq data.
- Sample-level boxplots: Per-sample median and interquartile range of log2 expression values across samples of 3 genes for readability were plotted to assess the outliers.
- Sample correlation heatmap: Pearson correlation matrix was visualized to assess the correlation between expressions of samples and cluster the data by similarity.
- Normality assessment via Q-Q plots and Shapiro-Wilk tests applied to randomly sampled genes.

3.3.3 Dimensionality Reduction

Prior to PCA, genes with zero variance across all samples were identified and removed using `apply(data_log, 1, var)`. PCA was then performed on this filtered matrix using the `prcomp()` function on transposed matrix (samples as rows). The first two principal components (PC1 and PC2) were extracted and visualized as scatter plots coloured by three clinical and biological variables: vital status (DECEASED vs. LIVING), gender (MALE vs. FEMALE), and sample type (Primary Tumor vs. Solid Tissue Normal vs. Recurrent Tumor). The percentage of variance explained by each component was calculated from the PCA importance matrix and displayed on the respective axis labels.

3.3.4 Differential Expression

Differential expression analysis was performed using the limma package on two-group comparison DECEASED versus LIVING (based on vital_status in the clinical metadata) , which implements linear modelling — the appropriate method for pre-normalized continuous expression data as DESeq2 / EdgeR was not used as it requires raw integer count matrices. Multiple testing correction was performed using the Benjamini-Hochberg (BH) procedure to control the false discovery rate (FDR).

Genes were declared differentially expressed if they met both criteria: (1) adjusted P-value < 0.05 and (2) |log2 fold change| > 0.5. Upregulated genes refer to higher expression in the DECEASED group relative to LIVING.

The results of DEGs are visualized by Volcano plot; log2 fold change (x-axis) vs. -log10(FDR) (y-axis). Upregulated genes shown in red, downregulated in blue, non-significant in grey

3.3.5 Functional Enrichment

Functional enrichment and pathway analysis were performed on the differentially expressed gene lists using the gprofiler2 package. Upregulated and downregulated DEGs were analyzed separately to distinguish biological processes enriched in each direction.

3.4 DNA Methylation Analysis

3.4.1 Data Acquisition and Preprocessing

DNA methylation analysis was performed using R statistical software (version 4.5.1). The analysis was initially applied to the project-provided subset of the TCGA-LIHC dataset; however, this subset yielded no statistically significant differentially methylated positions (DMPs), likely due to insufficient sample coverage and reduced statistical power. Consequently, the complete TCGA-LIHC Illumina HumanMethylation450 (450K) dataset was downloaded from the UCSC Xena GDC Hub (https://xenabrowser.net/datapages/?dataset=TCGA-LIHC.methylation450.tsv&host=https%3A%2F%2Fgdc.xenahubs.net) and all subsequent analyses were performed on this full dataset. Required R packages were loaded, including data.table, limma, minfi, IlluminaHumanMethylation450kanno.ilmn12.hg19, RColorBrewer, missMethyl, ggplot2, ggrepel, pheatmap, stringr, dplyr, and matrixStats. The Xena TSV file stores CpG probe identifiers in the first column, with TCGA sample barcodes as column headers and beta values (ranging from 0 to 1) as cell entries. To minimise memory consumption, column headers were read first to identify primary tumour samples, which were distinguished from adjacent normal samples by their tissue-code suffix (−01A or −01B). Only the tumour columns were loaded into memory using selective reading via fread.

Sample barcodes were standardised by removing the terminal vial letter (A or B), and duplicate barcodes arising from paired vials were resolved by retaining only the first occurrence. The resulting beta-value matrix was matched against clinical metadata, which were imported from tab-delimited files containing phenotypic variables and survival data. Shared sample identifiers between the methylation matrix and the clinical table were retained for downstream analyses. Key clinical covariates extracted included vital status (LIVING or DECEASED), gender, age at initial pathologic diagnosis, and pathologic stage, which was further simplified into binary Early (Stages I–II) and Late (Stages III–IV) categories.

3.4.2 Probe Filtering and Quality Control

Probes were subjected to a four-step sequential filtering procedure to ensure data quality and statistical validity prior to differential methylation analysis. First, probes with more than 20% missing values across samples were removed, as such probes represent unreliable measurements. Residual missing values in the remaining probes were imputed using the per-probe row median to satisfy the requirements of the downstream linear model. Second, probes mapping to sex chromosomes (chrX and chrY) were identified using the IlluminaHumanMethylation450kanno.ilmn12.hg19 annotation package and excluded to prevent confounding of methylation signals by differences in gender composition between groups. Third, probes containing single-nucleotide polymorphisms (SNPs) at either the CpG site or the single-base extension (SBE) position were removed, as sequence variation at these positions can artificially inflate differential methylation signals.

Data distribution was evaluated using global beta-value histograms and per-group density plots. The expected bimodal distribution of 450K array beta values, with peaks near 0 (unmethylated) and 1 (fully

methylated), was confirmed. Per-sample quality control was assessed using boxplots of a random 80-sample subset, and samples within each group displayed consistent median beta values, indicating acceptable array quality across the cohort.

3.4.3 Beta-to-M-Value Conversion

Beta values were converted to M-values before differential methylation analysis, as M-values exhibit more homoscedastic variance properties and are better suited to linear modelling. Beta values were first clamped to the interval (0.001, 0.999) to prevent infinite M-values at the distributional extremes, after which M-values were computed as:

$$M = log_2(\frac{\beta}{1-\beta})$$

3.4.4 Dimensionality Reduction

Principal Component Analysis (PCA) was performed on the top 5,000 most variable CpG probes, selected by row variance across the M-value matrix, to explore global methylation structure across samples. Six pairwise combinations of the first four principal components (PC1–PC4) were visualised and coloured according to four clinical variables: vital status, gender, pathologic stage, and age at diagnosis.

Additionally, Multi-Dimensional Scaling (MDS) was performed using the plotMDS function from the limma package on the top 1,000 most variable probes, providing a complementary dimensionality reduction view coloured by vital status and gender. Hierarchical clustering heatmaps were generated for the top 1,000 and top 50 most variable CpGs using row-scaled M-values, with columns annotated by vital status, gender, and pathologic stage.

3.4.5 Differential Methylation Analysis

Differential methylation analysis between DECEASED and LIVING patient groups was performed using the limma linear modelling framework, which is widely used for microarray-based methylation data. Samples with missing vital status were excluded prior to model fitting. A design matrix was constructed using a no-intercept parameterisation that included vital status and gender as covariates, the latter included to account for residual sex-linked methylation variation.

Probes were considered significantly differentially methylated when they satisfied both of the following criteria:

- Adjusted P-value (Benjamini–Hochberg FDR) < 0.05
- Absolute log-fold change in M-value (|logFC|) > 0.5

Results were visualised using a volcano plot displaying the M-value log-fold change (DECEASED - LIVING) against the $-\log_{10}$ raw P-value. Hypermethylated probes in DECEASED patients (logFC > +0.5) were coloured in red, hypomethylated probes (logFC < −0.5) in blue, and non-significant probes in grey. The top 20 significant probes were labelled with their associated gene names. Individual probe-level beta-value distributions for the six most significant DMPs were visualised as jittered boxplots. The genomic distribution of significant DMPs was further characterised by their CpG island context (Island, Shore, Shelf,

Open Sea) and chromosomal location. Full results and the filtered significant DMP table were exported as CSV files for downstream analyses.

3.4.6 Gene Ontology and Pathway Enrichment

Gene Ontology (GO) and KEGG pathway enrichment analyses were performed on the genes associated with significant DMPs using the gometh function from the missMethyl package. This function applies a bias correction accounting for the unequal number of 450K probes per gene, which would otherwise inflate enrichment signals for highly covered genes. The background set comprised all probes retained after quality filtering. GO terms spanning Biological Process (BP), Molecular Function (MF), and Cellular Component (CC) ontologies were tested, alongside KEGG pathways. Statistical significance was assessed using the Benjamini–Hochberg FDR, with enriched terms considered significant at FDR < 0.05. Results were visualised as horizontal bar charts plotting the number of differentially expressed genes per term coloured by FDR, and as bubble plots displaying $-\log_{10}(P)$ on the x-axis with point size scaled to the number of associated genes.

3.5 miRNA Analysis

3.5.1 Data Acquisition and Preprocessing

MicroRNA (miRNA) expression data and the corresponding clinical metadata were analysed using the R statistical software (version 4.5.1). Required packages were loaded, including DESeq2, NMF, readxl, ggplot2, and pheatmap.

```
library(DESeq2)
library(readxl)
library(ggplot2)
library(pheatmap)
```

The miRNA expression matrix was imported as a numeric matrix, while the clinical metadata were imported from an Excel spreadsheet.

```
data <- as.matrix(read.table("mirna", header = TRUE,
                             row.names = 1))
clinical <- as.data.frame(read_excel(file.choose()))
```

To ensure consistency between datasets, clinical sample identifiers were reformatted by replacing hyphens with periods. Samples present in both datasets were identified and retained for subsequent analyses.

```
clinical$sampleID_dot <- gsub("-", ".", clinical$sampleID)

common_samples <- intersect(colnames(data),
                            clinical$sampleID_dot)

mirna_clean <- data[, common_samples]
clinical_clean <- clinical[
    match(common_samples,
          clinical$sampleID_dot), ]
```

Sample alignment was verified prior to downstream analyses.

```
all(colnames(mirna_clean) ==
    clinical_clean$sampleID_dot)
```

Expression distributions were evaluated using histograms, boxplots, quantile–quantile (Q–Q) plots and the Shapiro–Wilk normality test. To reduce skewness and stabilize variance, expression values were transformed using the $\log_2(x + 1)$ transformation.

```
mirna_log <- log2(mirna_clean + 1)

hist(mirna_log)

boxplot(log2(mirna_clean[1:5,] + 1))

qqnorm(mirna_clean[3,])
qqline(mirna_clean[3,])

shapiro.test(mirna_clean[100,])
```

A heatmap of the 50 most variable miRNAs was generated following variance filtering to visualize global expression patterns across samples.

```
row_variances <- apply(mirna_log, 1, var)

top50 <- names(sort(row_variances,
                    decreasing = TRUE))[1:50]

pheatmap(mirna_log[top50, ],
          scale = "row",
          annotation_col = annotation)
```

3.5.2 Principal Component Analysis

Principal Component Analysis (PCA) was performed using the log-transformed miRNA expression matrix after removing miRNAs with zero variance. Expression values were standardized before PCA, and the first two principal components were used to evaluate sample clustering.

```
row_variances <- apply(mirna_log, 1, var)

variable_mirnas <- mirna_log[
    row_variances > 0, ]

pca_result <- prcomp(
    t(variable_mirnas),
    scale. = TRUE
)
```

Samples were coloured according to clinical characteristics, including **vital status** and **gender**, to investigate whether these variables influenced global miRNA expression patterns.

```
ggplot(pca_df,
       aes(PC1, PC2,
           color = Vital_Status)) +
    geom_point()

ggplot(pca_df,
       aes(PC1, PC2,
           color = Gender)) +
    geom_point()
```

3.5.3 Differential Expression

Differential expression analysis was performed using the DESeq2 Bioconductor package. Patient samples were classified into LIVING and DECEASED groups according to the clinical variable *vital_status*, with LIVING designated as the reference group.

```
coldata$vital_status <-
    relevel(
        factor(clinical_clean$vital_status),
        ref = "LIVING"
)
```

A DESeq2 dataset was created from the raw miRNA count matrix. Lowly expressed miRNAs (total count < 10) were excluded prior to model fitting.

```
dds <- DESeqDataSetFromMatrix(
    countData = round(mirna_clean),
    colData = coldata,
    design = ~ vital_status
)

dds <- dds[
    rowSums(counts(dds)) >= 10, ]
```

Differential expression was estimated using the DESeq2 negative binomial generalized linear model.

```
dds <- DESeq(dds)

res <- results(dds)
```

Log_2 fold changes and Benjamini–Hochberg adjusted p-values (FDR) were extracted for each miRNA. MiRNAs were considered significantly differentially expressed when they satisfied both of the following criteria:

- Adjusted *P* value (FDR) < 0.05
- Absolute log_2 fold change > 0.5

```
significant_mirnas <-
    subset(de_results,
```

```
            Padj < 0.05 &
            abs(Log2FC) > 0.5)
```

The differential expression results were visualized using a volcano plot displaying $\log_2$ fold change against $-\log_{10}$(adjusted *P*-value), and significant miRNAs were exported as a CSV file for downstream analyses.

```
ggplot(de_results,
       aes(Log2FC,
           -log10(Padj),
           color = Significance)) +
    geom_point()

write.csv(significant_mirnas,
          "significant_differential_miRNAs_DESeq2.csv",
          row.names = FALSE)
```

3.5.4 Functional Enrichment

Significant differentially expressed miRNAs were subjected to functional enrichment analysis using the miEAA web server. Over-representation analysis (ORA) was performed using Homo sapiens as the reference organism. Enriched Gene Ontology (GO), KEGG, Reactome, and disease-associated categories were identified based on experimentally validated and predicted miRNA target annotations. Pathways with an adjusted P-value < 0.05 were considered significantly enriched.

3.6 MOFA Integration Analysis

3.6.1. Multi-Omics Factor Analysis Framework

Multi-Omics Factor Analysis (MOFA+) was applied to identify latent axes of variation across multi-modal molecular datasets capturing heterogeneous biological signals. The analysis was performed on a highly matched retrospective cohort comprising N = 410 samples with complete overlap across omics layers, providing sufficient statistical power for unsupervised factor inference.

All computational analyses were conducted in R (v4.x) using the `MOFA2` framework alongside supporting packages such as `data.table`.

3.6.2. Preprocessing and Data Transformation

MOFA+ assumes continuous data modeled under a Gaussian likelihood; therefore, all input data were transformed to approximate normal distributions and ensure numerical stability.

DNA methylation preprocessing

DNA methylation β-values were first adjusted to avoid boundary issues during logit transformation. Values equal to 0 or 1 were replaced with a small offset (± 10^-6). The adjusted β-values were then converted into M-values using:

$$M=\log2(\beta1-\beta)$$

Expression data preprocessing

mRNA and miRNA expression values were normalized and log-transformed using:

$$X'=\log2(X+1)$$

This transformation reduces skewness and stabilizes variance across samples.

3.6.3. High-Variance Feature Selection

To reduce dimensionality and improve signal-to-noise ratio, feature selection was performed independently for each omics layer based on sample-wise variance.

To address imbalance in feature dimensionality across modalities, stricter filtering was applied to high-dimensional views. The top variable features were retained as follows:

- mRNA: top 5,000 highly variable genes
- DNA methylation: top 5,000 highly variable CpG sites
- miRNA: top 500 highly variable transcripts

This step improves computational efficiency and enhances latent factor interpretability.

3.6.4. MOFA Object Construction

Preprocessed datasets were organized into a multi-view structure, where rows represent features and columns represent matched samples across all modalities.

```
data_list <- list(
  mRNA        = mRNA_data,
  Methylation = methylation_data,
  miRNA       = miRNA_data
)

MOFAobject <- create_mofa(data_list)
```

Model consistency and sample alignment were verified using MOFA diagnostic functions (e.g., data overview plots).

3.6.5. Model Configuration and Training

Model configuration was defined using separate option objects controlling data scaling, likelihood assumptions, and optimization strategy.

- Data scaling was enabled (`scale_views = TRUE`) to normalize variance across modalities.
- A Gaussian likelihood was assigned to all views (`"gaussian"`).
- Model optimization was performed using variational inference with a conservative convergence setting (`convergence_mode = "slow"`).
- Maximum iterations were set to 10,000, and a fixed seed (42) was used to ensure reproducibility.

The model was trained and saved using:

```
MOFAobject <- prepare_mofa(
  object           = MOFAobject,
  data_options     = data_opts,
  model_options    = model_opts,
  training_options = train_opts
)

outfile <- file.path(getwd(), "TCGA_MOFA_model.hdf5")
MOFAobject <- run_mofa(MOFAobject, outfile = outfile)
```

3.6.6. Latent Space Integration and Clinical Annotation

The trained model was imported along with clinical metadata. Sample identifiers were standardized by harmonizing formatting differences (hyphens replaced with periods) to ensure consistent mapping across datasets.

Samples were restricted to the intersecting cohort using MOFA subsetting functions.

Overall survival variables were constructed using:

- `days_to_death` for deceased patients
- `days_to_last_followup` for censored observations

This enabled time-to-event survival modeling across the integrated cohort.

3.6.7. Latent Factor Analysis

Latent factors were extracted using MOFA's internal factor retrieval functions. Variance explained (R^2) was computed to quantify contributions of each factor across omics layers.

Dimensional structure was visualized using Uniform Manifold Approximation and Projection (UMAP) applied to latent factors.

Associations between latent factors and clinical variables (age, gender, tumor stage, and survival status) were assessed using correlation-based analyses.

3.6.8. Survival Modeling

The prognostic relevance of latent factors was evaluated using Cox proportional hazards regression. Hazard ratios and Wald test p-values were computed for each factor.

For visualization, patients were stratified into high- and low-risk groups based on median factor values, and Kaplan–Meier survival curves were generated.

3.6.9. Feature Extraction and Functional Enrichment

Feature loadings were extracted from the MOFA model to identify molecular contributors to each latent factor. Within the transcriptomic view, features were ranked by absolute weight values, and the top 300 genes per factor were selected for downstream analysis.

Gene identifiers were cleaned to retain standardized HUGO symbols and converted to Entrez IDs using the `clusterProfiler` framework with the `org.Hs.eg.db` annotation database.

Functional enrichment analysis was performed for each factor using Gene Ontology (Biological Process) and KEGG pathway databases. Multiple testing correction was applied using the Benjamini–Hochberg method.

Enrichment analyses were conducted in batch mode across the top 15 latent factors to systematically characterize factor-specific biological programs.

4. Results

4.1 RNA-seq Results

4.1.1 Expression Distribution and Quality Control

The histogram of mean log2(FPKM+1) values per gene (Figure 1) exhibited the characteristic bimodal distribution of RNA-seq data, with a prominent peak near zero for unexpressed genes and a broader peak of expressed genes centered around log2 values.

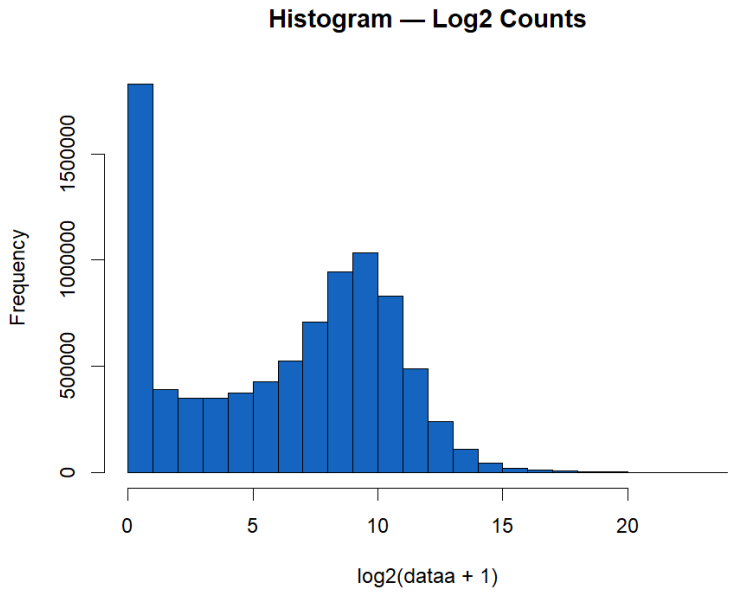


Figure 1:

Per-sample boxplots of log2(FPKM+1) distributions across samples of 3 genes for readability demonstrated comparable median expression levels and interquartile ranges.

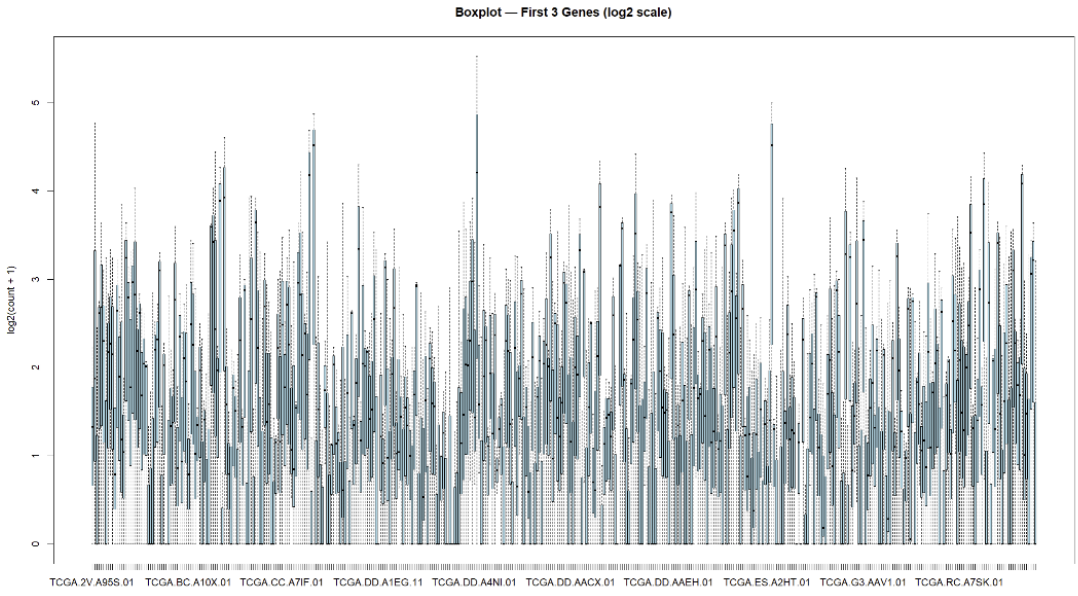
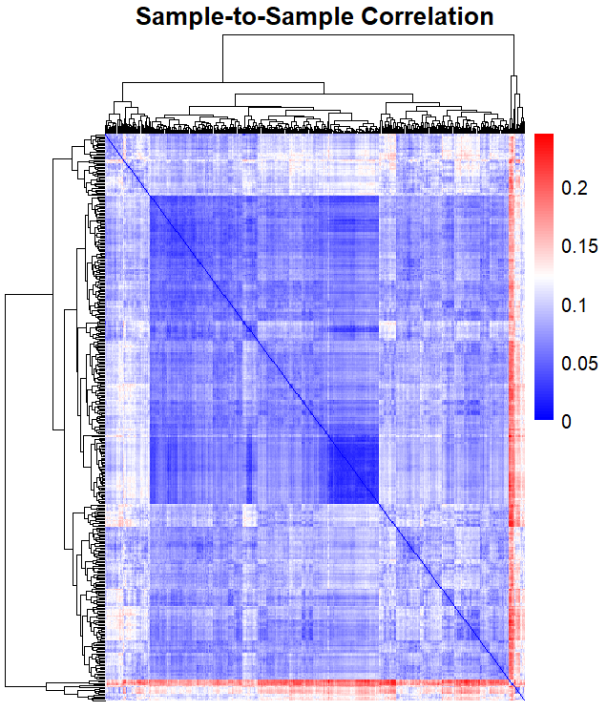


Figure 2:

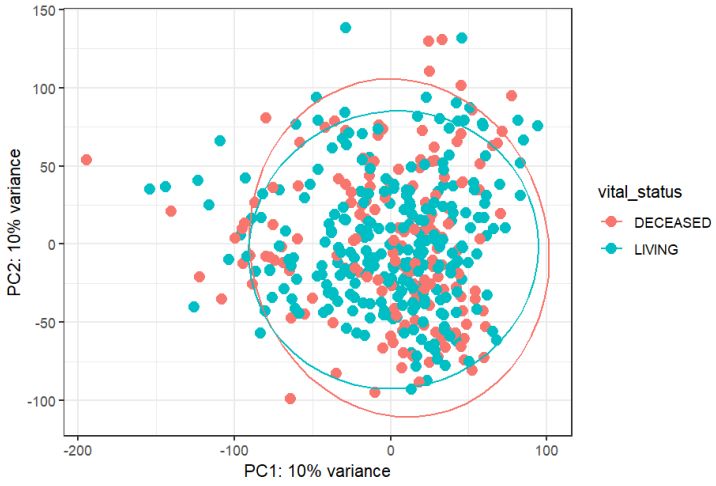
Sample-to-Sample Correlation: The Pearson correlation heatmap of genes (Figure 3) revealed high overall correlation among samples. There is a large blue block and distinct red regions at the edges. The red regions visible at the periphery of the heatmap represent samples with low correlation to the main cluster.



4.1.2 Principal Component Analysis

PCA — Vital Status (Figure 4)

PCA of the log2-normalized expression matrix revealed that the first two principal components each explained 10% of total transcriptomic variance, together capturing 20% of the total variation across genes. Critically, coloring samples by vital status (DECEASED vs. LIVING) showed complete overlap between the two groups — neither PC1 nor PC2 separated patients by survival outcome. This finding indicates that vital status is not the dominant driver of global expression variation in this cohort. The PCA showed no variation between DECEASED and LIVING patients.



PCA — Gender

Coloring the same PCA projection by patient gender revealed no separation between male and female samples along PC1 or PC2. The confidence ellipses for both groups were largely coincident, indicating that gender does not drive major global transcriptomic differences in this HCC cohort.

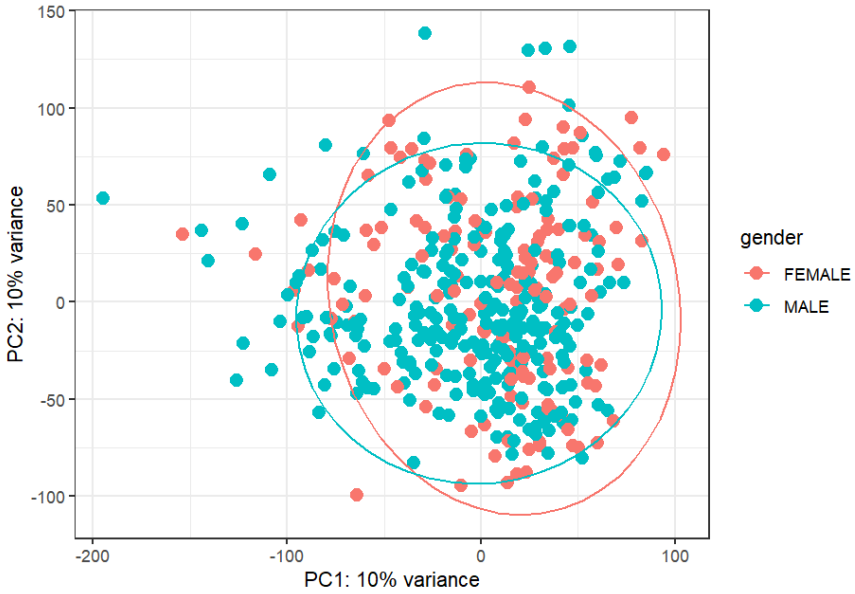


Figure 5:

PCA — Sample Type

The PCA colored by sample type revealed the most biologically informative separation in the dataset. Adjacent-normal solid tissue samples (Solid Tissue Normal, n=50) formed a tight, distinct cluster in the lower portion of the PC1/PC2 space, clearly separated from the primary tumor samples. This separation confirms that the major axis of transcriptomic variation in the dataset is the biological difference between tumor and non-tumor hepatic tissue.

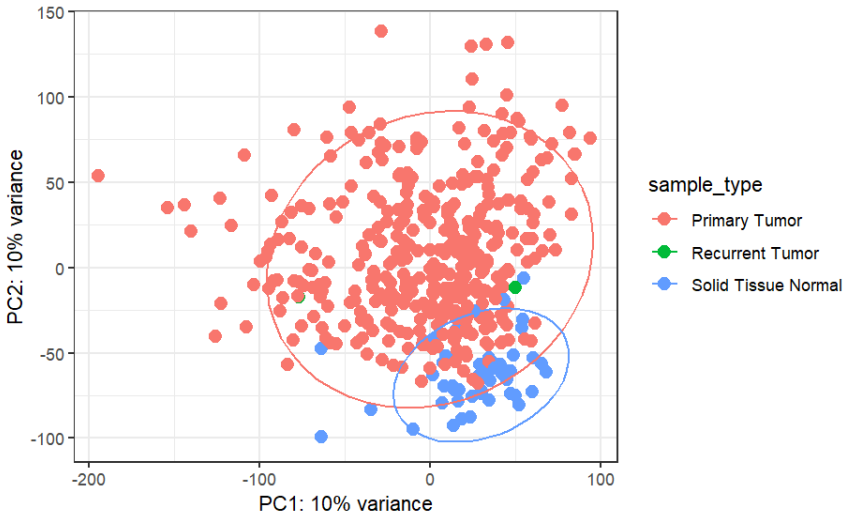
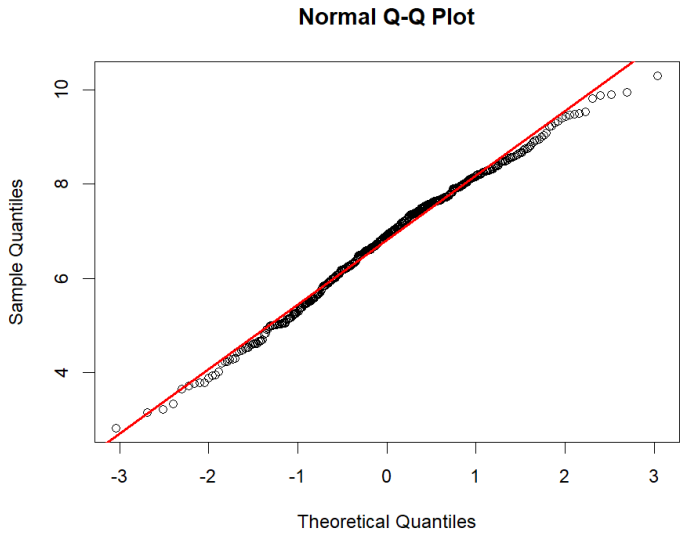


Figure 6:

4.1.3 Normality Assessment

Q-Q plots for randomly selected genes Figure 7 showed that log2(FPKM+1) values not normally distributed. The Shapiro-Wilk test applied giving p-value = 0.004, less than 0.05 means the normality is rejected. The results shows the data is not normally distributed.

Figure 7:



4.1.4 Differential Expression

DEGs

Differential expression analysis comparing DECEASED versus LIVING identified statistically significant differentially expressed genes (DEGs) at adj.P.Val < 0.05 and |log2FC| > 0.5 (Table 1).

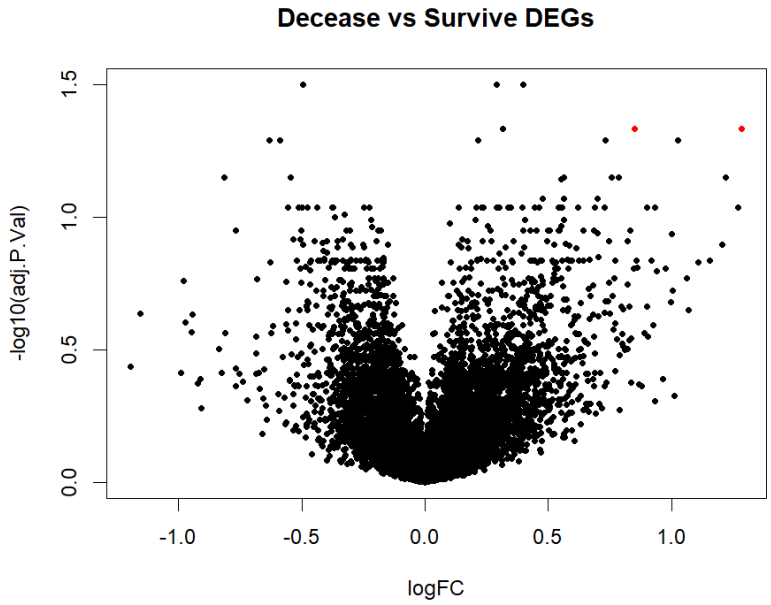
Differential expression analysis identified two statistically significant genes (adj.P.Val < 0.05, |log2FC| > 0.5), both upregulated in DECEASED relative to LIVING patients. The first gene, **EPO** (Erythropoietin, logFC = +1.29), encodes a cytokine classically associated with erythropoiesis but increasingly recognized as a pro-tumorigenic factor in HCC, promoting tumor cell survival and resistance to apoptosis. The second gene, **S100A9** (S100 Calcium Binding Protein A9, logFC = +0.85), is a damage-associated molecular pattern protein involved in inflammatory signaling, neutrophil recruitment, and tumor microenvironment remodeling. While the small number of DEGs limits pathway-level interpretation, both identified genes are biologically coherent with aggressive HCC phenotypes and worse survival outcomes, supporting the analytical direction.

Gene	$\log_2$FC	AveExpr	t	P-value	adj.P
EPO	+1.286	3.810	4.440	1.15×10^{-5}	0.0464
S100A9	+0.851	7.302	4.399	1.38×10^{-5}	0.0464

Volcano Plot

The volcano plot of all tested genes showed a compressed distribution with a narrow range of fold changes (logFC approximately −1.0 to +1.3) and low statistical significance across genes (-log10(adjusted P-value) < 1.5). Only two genes reached statistical significance at adjusted P-value < 0.05 with |logFC| > 0.5, both upregulated in DECEASED patients. The limited number of DEGs and the compressed volcano shape are attributable to the inclusion of adjacent-normal samples in the analysis cohort, which introduces a large source of expression variation unrelated to vital status.

Figure 8:



4.1.4 Functional Enrichment

Functional enrichment and pathway analysis were performed on the upregulated and downregulated DEG lists identified by the limma analysis. However, no statistically significant enriched terms were returned by g:Profiler for either the upregulated or downregulated gene sets across any of the queried databases (GO:BP, GO:MF, GO:CC, KEGG, or Reactome).

This outcome is a direct consequence of the limited DEG yield from the differential expression analysis. Only two genes satisfied the significance criteria (FDR < 0.05, |log2FC| > 0.5): EPO (Erythropoietin, log2FC = +1.29, FDR = 0.046) and S100A9 (S100 Calcium Binding Protein A9, log2FC = +0.85, FDR = 0.046), both upregulated in the DECEASED group. No genes were identified in the downregulated direction. Over-representation analysis requires a sufficiently large gene list to detect statistically meaningful enrichment of biological terms.

As a result, no enrichment plots, pathway maps, or GO term tables could be generated from the current analysis. The gprofiler2 result objects (gp_up and gp_down) returned empty result tables confirming the absence of significant terms.

analysis requires a sufficiently large gene list to detect statistically meaningful enrichment. A refined analysis restricted to primary tumor samples only is expected to yield a larger DEG set enabling full pathway interpretation.

4.2 DNA Methylation Results

4.2.1 Data Acquisition and Quality Control

The initial analysis was performed on the project-provided TCGA-LIHC methylation subset; however, this dataset yielded no statistically significant differentially methylated positions (DMPs) upon application of the full analysis pipeline, most likely owing to the limited sample size and the consequently reduced statistical power of the differential testing procedure. To address this limitation, the complete TCGA-LIHC Illumina HumanMethylation450 (450K) DNA methylation dataset was retrieved from the UCSC Xena GDC Hub. Following selective loading of primary tumour samples (tissue-code suffix −01A or −01B), the raw beta-value matrix comprised **486427** CpG probes across **377** samples.

The four-step probe filtering procedure progressively reduced the number of probes to a high-confidence set suitable for statistical modelling. Probes with more than 20% missing values were first removed, retaining 394675 probes, after which residual missing values were imputed using the per-probe row median. Removal of sex-chromosome probes (chrX and chrY) further reduced the probe count to **385952**, preventing any residual gender-linked methylation from confounding the survival-based comparison. Subsequent exclusion of probes harbouring SNPs at the CpG site or single-base extension position yielded a final filtered set of **319133** probes. Following clinical matching, **377** tumour samples were retained with complete vital status annotation, comprising **132** DECEASED and **245** LIVING patients.

Examination of the global beta-value distribution confirmed the characteristic bimodal profile expected from Illumina 450K array data, with the two peaks near 0 and 1 corresponding to unmethylated and fully methylated CpG sites, respectively, Figure X. Per-group density plots demonstrated broadly overlapping distributions between DECEASED and LIVING patients on both the beta and M-value scales, suggesting that global methylation levels are largely comparable between the two groups at the array-wide level. Per-sample quality control boxplots, generated for a random subset of 80 samples, revealed consistent median beta values and interquartile ranges across all samples, confirming acceptable and homogeneous array quality with no obvious outlier samples requiring exclusion, Figure 9.

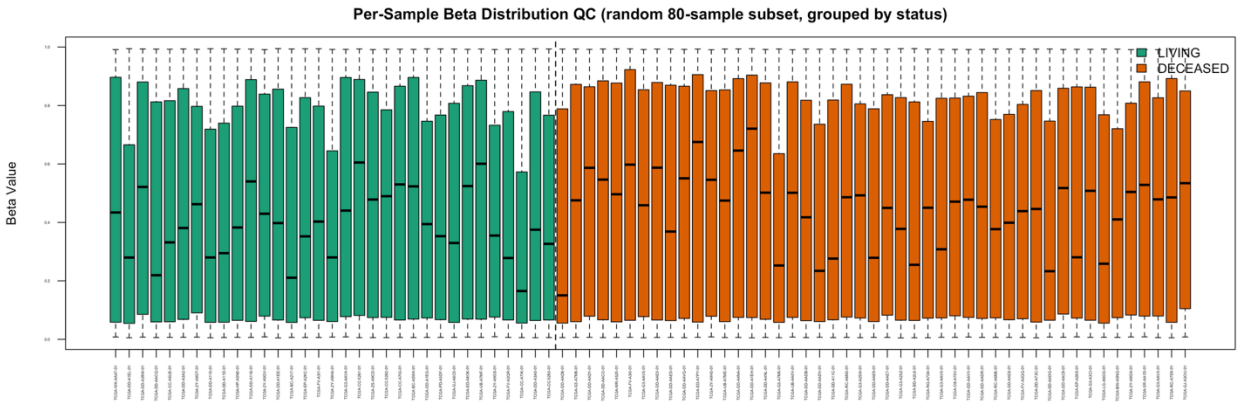


Figure 9: Per-sample beta distribution plot for deceased and living samples.

4.2.2 Exploratory Analysis: Heatmaps and Variance Structure

Hierarchical clustering heatmaps were constructed using the top 1,000 and top 50 most variable CpG probes, ranked by row variance across the M-value matrix. The top-1,000 heatmap revealed considerable heterogeneity in methylation profiles across the patient cohort, with samples annotated by vital status, gender, and pathologic stage, Figure 10. Although distinct probe clusters exhibiting coordinate hyper- or hypomethylation were apparent, no clear macroscopic separation of DECEASED and LIVING patients was observed across the columns. This pattern was consistently reproduced in the top-50 heatmap, where individual probe identifiers were visible, Figure 11. The absence of gross sample-level clustering by survival status at this exploratory stage is consistent with observations from the global miRNA expression analysis and indicates that survival-associated methylation differences are likely restricted to a subset of loci rather than representing a global epigenomic shift detectable by unsupervised methods.

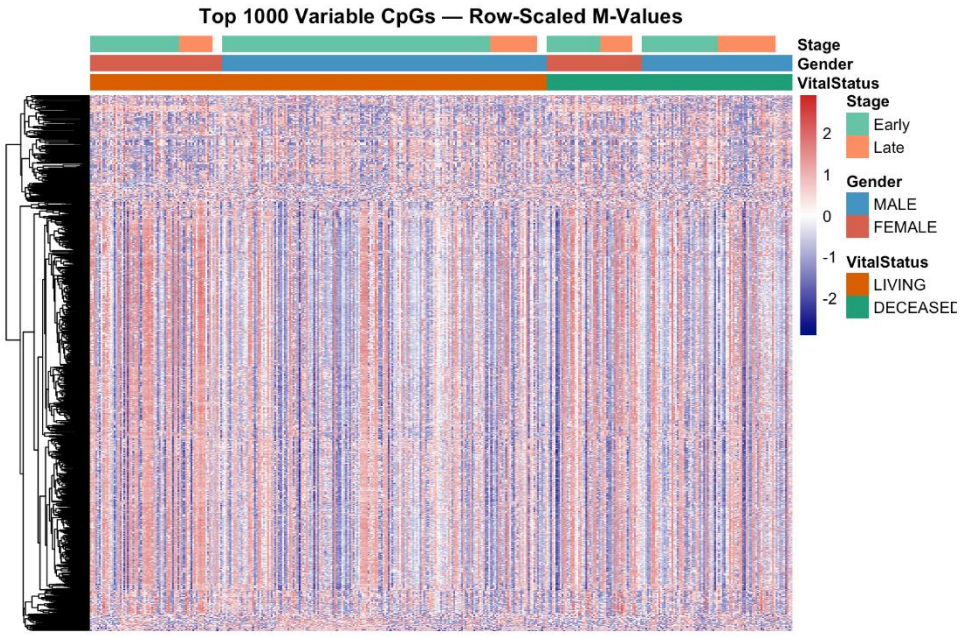


Figure 10: Top 1000 variable CpGs – Row Scaled M-values.

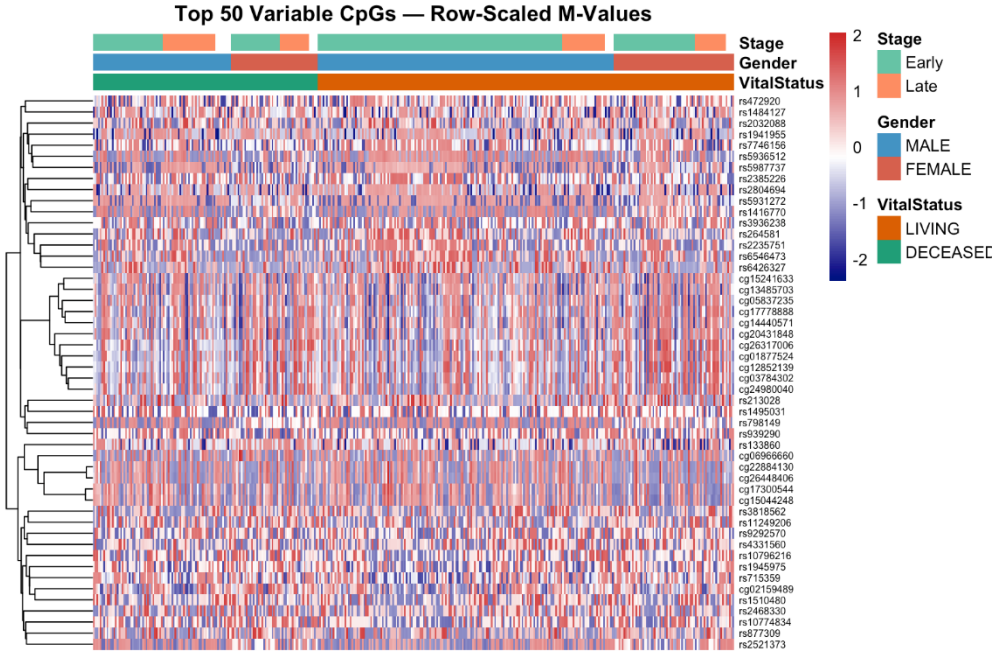


Figure 11: Top 50 variable CpGs – Row Scaled M-values.

4.2.3 Principal Component Analysis and MDS

Principal Component Analysis was performed on the top 5,000 most variable CpG probes to characterise the dominant sources of methylation variance across the cohort. The first four principal components collectively explained **64.5%** of total variance, with PC1 accounting for **55**%, PC2 for **5.4%,** PC3 for **2.5**%, and PC4 for **1.6%,** as depicted in PCA for vital status, Figure 12. Six pairwise combinations of PC1–PC4 were visualised and coloured according to vital status, gender, pathologic stage, and age at diagnosis.

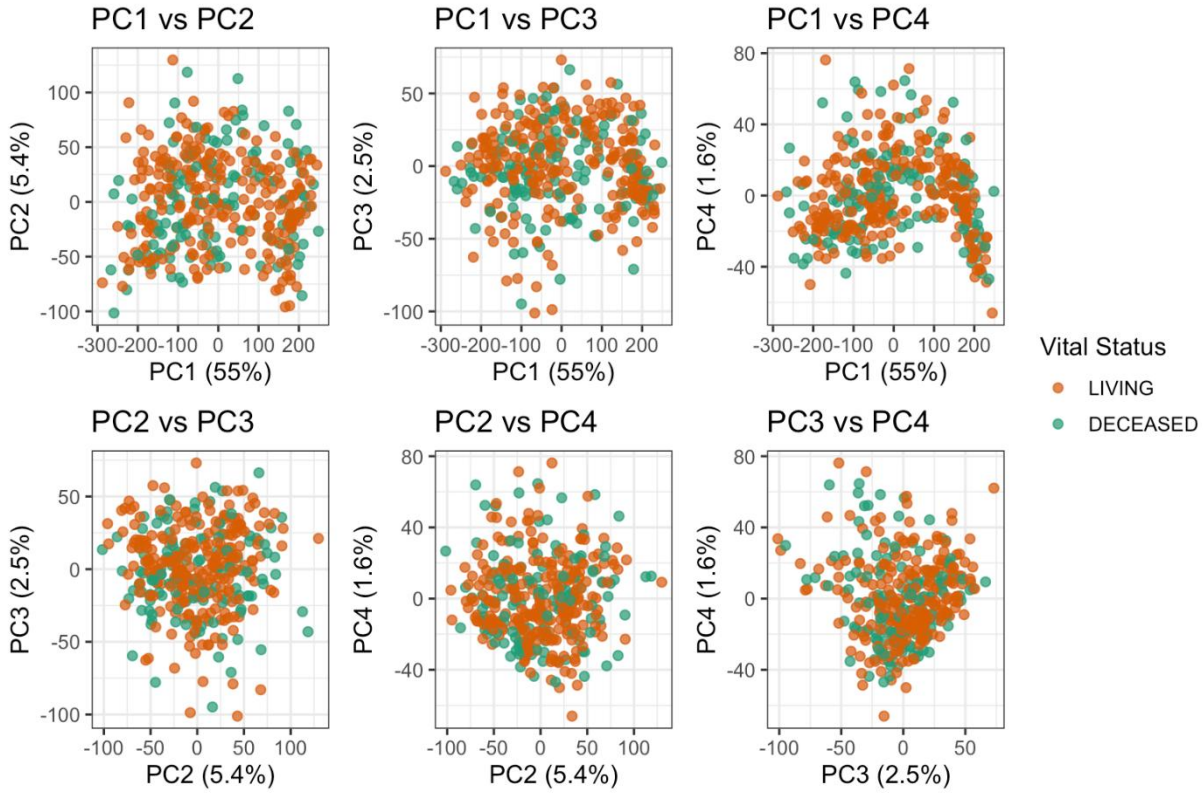


Figure 12: Principal Component Analysis (PCA) for vital status.

When samples were coloured by vital status, no discrete separation between DECEASED and LIVING patients was observed across any of the PC pair projections, with the two groups exhibiting substantial coordinate overlap throughout the PCA space, Figure 4. Similarly, colouring by gender did not reveal structured clustering along any principal component, confirming that the removal of sex-chromosome probes during quality control was effective in eliminating gender-driven methylation variance. No structured gradient was discernible when samples were coloured by continuous age at diagnosis.

Multi-Dimensional Scaling (MDS), performed using limma on the top 1,000 most variable probes, produced results consistent with the PCA findings, with DECEASED and LIVING samples intermixing extensively in the two-dimensional MDS plane and no sharp boundary separating the two groups by vital status or gender. Collectively, these exploratory results demonstrate that the leading axes of methylation variance across the TCGA-LIHC cohort are not predominantly driven by patient survival status or

demographic variables. This convergent finding across two independent omics layers reinforces the conclusion that survival-associated molecular differences are low-amplitude, localised signals that require supervised statistical testing for their identification.

4.2.4 Differential Methylation Analysis

Differential methylation analysis was conducted using the limma linear modelling framework applied to the M-value matrix. Gender was included as a covariate in the design matrix to account for any residual sex-linked methylation variability not fully eliminated during probe filtering. Empirical Bayes variance shrinkage was applied following contrast extraction of the DECEASED versus LIVING comparison.

Probes were classified as significantly differentially methylated when they satisfied both of the following criteria:

- Adjusted P-value (Benjamini–Hochberg FDR) < 0.05
- Absolute M-value log-fold change (|logFC|) > 0.5

The analysis identified **401** significant DMPs in total, of which **39** were hypermethylated in DECEASED patients (logFC > +0.5) and **362** were hypomethylated in DECEASED patients (logFC < −0.5). The delta-beta (Δβ) values for significant DMPs ranged from **-0.316 to 0.176,** confirming that the identified probes exhibited not only statistically significant but also biologically meaningful differences in methylation level between the two groups. The raw P-value histogram displayed a distribution enriched toward zero with a broadly uniform right tail, Figure 5, confirming the presence of a genuine differential methylation signal and validating the suitability of the limma model for this dataset.

Hypermethylated DMPs	Hypomethylated DMPs	Total Significant DMPs
39 (DECEASED)	**362** (DECEASED)	**401** FDR < 0.05

Table 3. Summary of differentially methylated positions (DMPs).

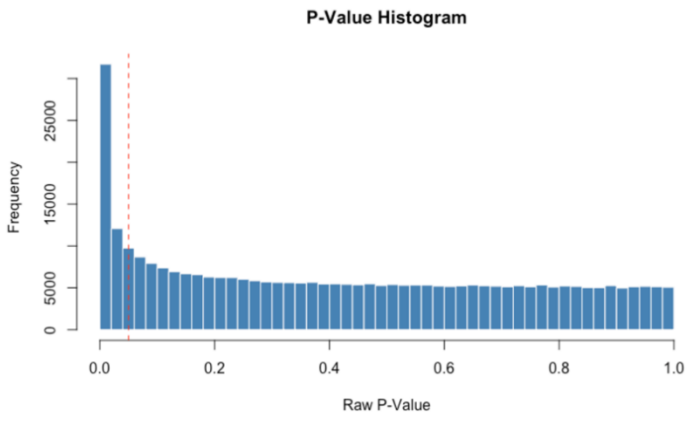


Figure 13: Histogram showing the raw p-value of the total methylation probes.

Among the most significantly hypermethylated CpG probes in DECEASED patients were **cg15421984**, while the most significantly hypomethylated probes included **cg12648336**, Figure 6. The full results table and the filtered significant DMP table were exported as CSV files for downstream enrichment analyses.

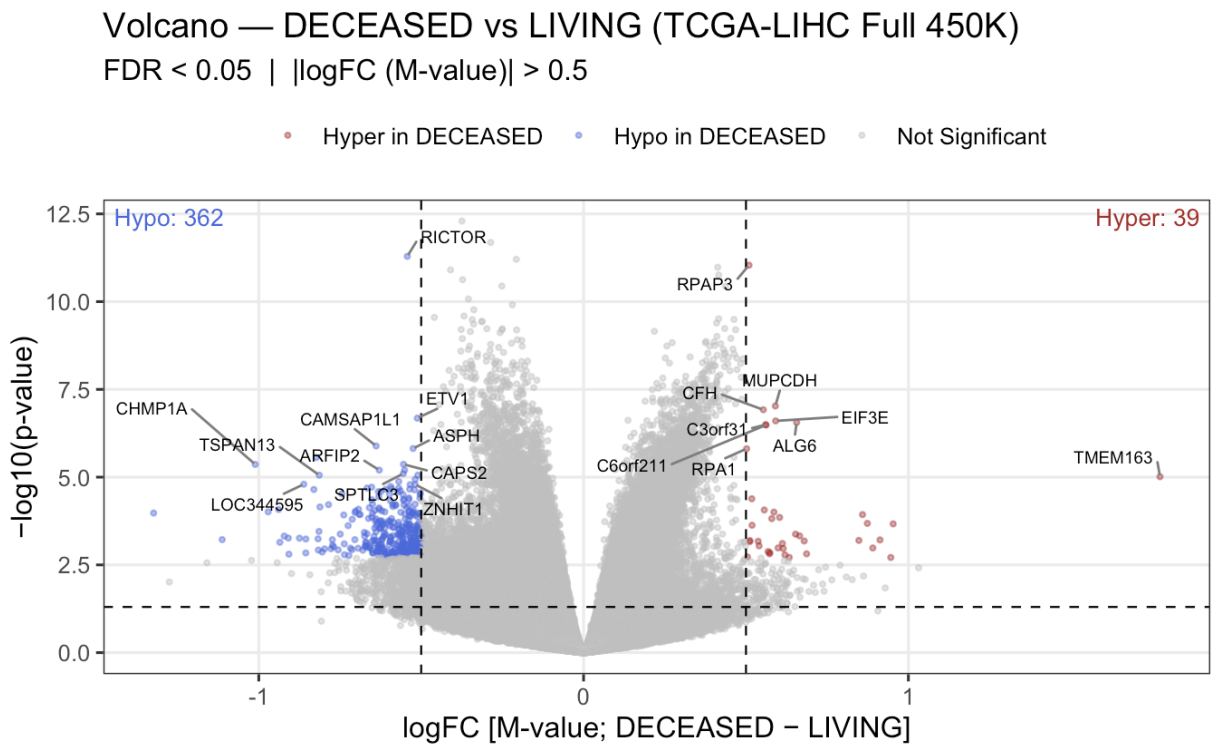


Figure 14: Volcano plot showing hyper and hypomethylated DMPs.

4.2.5 Genomic Distribution of Significant DMPs

The chromosomal distribution of significant DMPs revealed a non-uniform spread across the genome, with enrichment on chromosomes 1,2, and 3, Figure 15. Enrichment on specific chromosomes may reflect the presence of epigenetic hotspots or genomic regions particularly susceptible to methylation dysregulation in hepatocellular carcinoma.

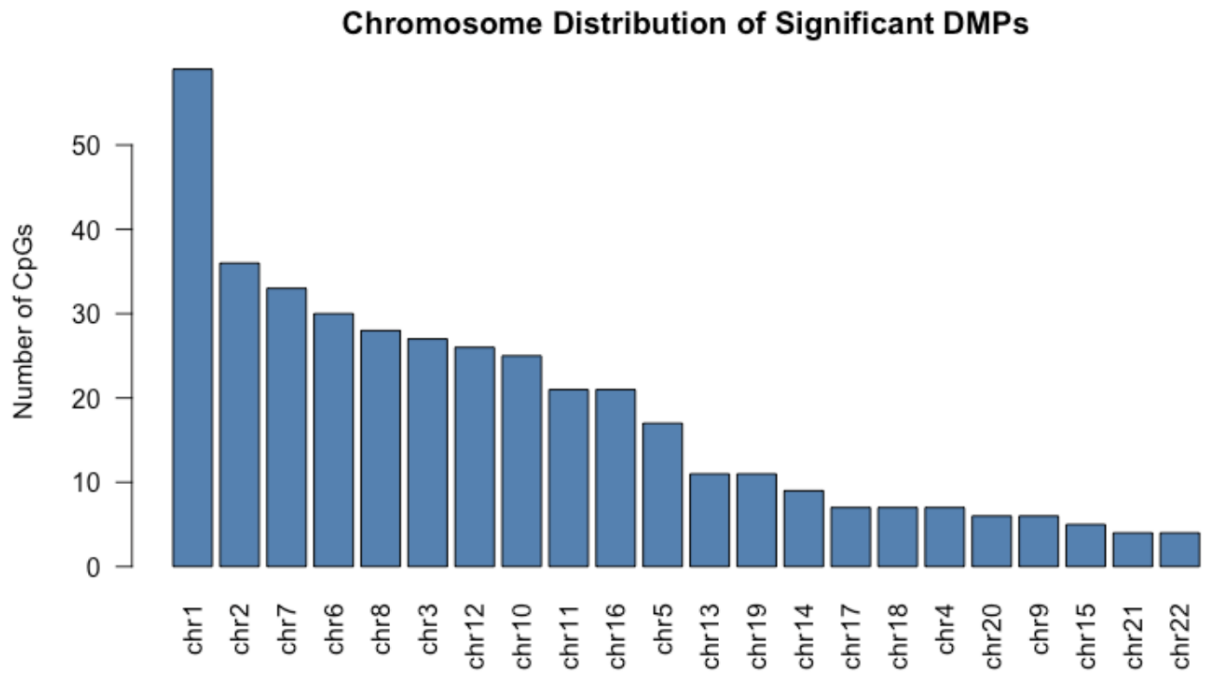


Figure 15: Chromosome Distribution of Significant DMPs.

Analysis of the CpG island context of significant DMPs demonstrated that the majority of hypermethylated probes in DECEASED patients were located in CpG **OpenSea**, while hypomethylated probes were more frequently found in **Island** regions. Hypermethylation of CpG island-associated promoter-proximal regions is a well-established mechanism of tumour suppressor gene silencing in cancer, whereas hypomethylation of distal intergenic or repeat-element CpG sites can promote genomic instability and aberrant oncogene activation. The observed distribution of DMPs across these genomic contexts is therefore consistent with broad patterns of epigenetic dysregulation previously described in hepatocellular carcinoma.

4.2.6 Gene Ontology and Pathway Enrichment

To characterise the biological processes associated with the [X] significant DMPs, the genes mapped to these probes were subjected to bias-corrected gene set enrichment analysis using the gometh function from the missMethyl package, which adjusts for the unequal number of 450K array probes per gene. Over-representation analysis was performed against Gene Ontology (GO) and KEGG pathway databases, with statistical significance assessed at FDR < 0.05.

GO enrichment analysis identified 15 significant terms spanning Biological Process, Molecular Function, and Cellular Component ontologies, Figure 16. The most significantly enriched Biological Process terms included **[e.g., actin-mediated cell contraction, regulation of ventricular cardiac muscle cell action potential]**, suggesting that the differentially methylated loci collectively target fundamental cellular regulatory programmes relevant to cancer progression. At the Molecular Function level, enrichment was observed for terms related to **[e.g., 2-oxoglutarate-dependent dioxygenase activity],** which is consistent with the propensity of CpG island methylation to regulate transcription factor accessibility at gene promoters.

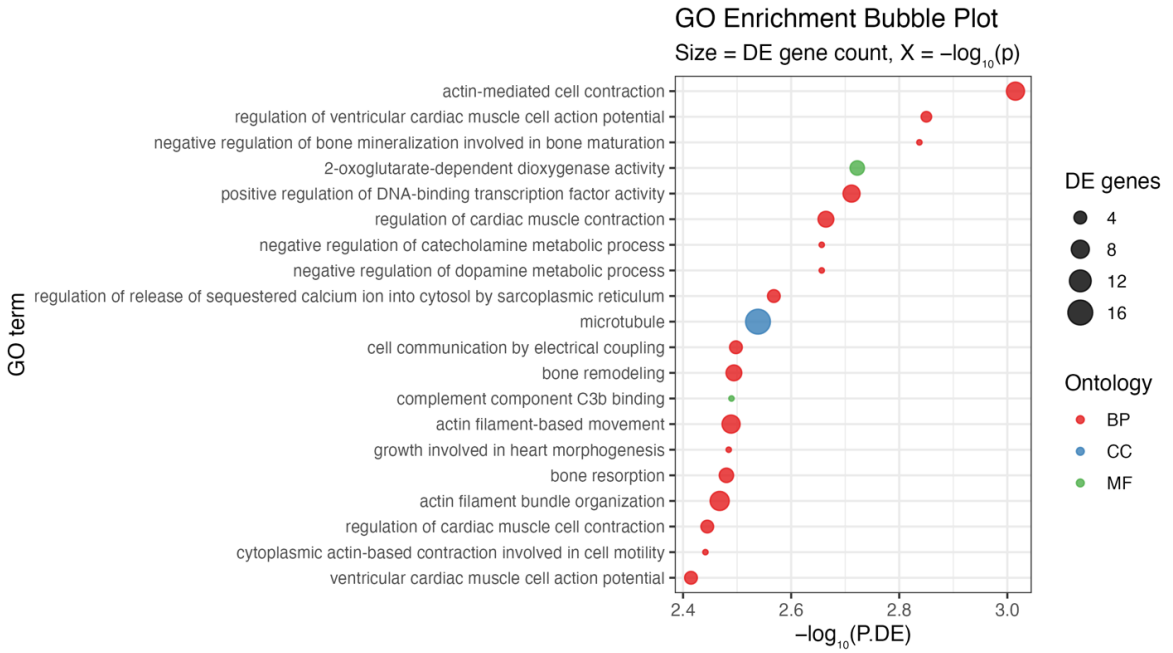


Figure 16: A bubble plot showing the top 15 significant GO enrichment.

KEGG pathway enrichment analysis identified 15 significantly enriched pathways, Figure 17. Prominent pathways with well-established roles in hepatocellular carcinoma biology, including regulation of cell proliferation, survival signalling, and immune evasion. These findings are concordant with the pathway-level dysregulation observed in the miRNA enrichment analysis and point toward convergent epigenetic and post-transcriptional regulation of shared oncogenic programmes in the DECEASED patient group. Taken together, the differential methylation results demonstrate that although global methylation profiles do not separate patient groups at the unsupervised level, a distinct set of loci undergoes significant and biologically coherent methylation changes in association with patient survival, providing a complementary epigenomic layer of evidence to the miRNA expression findings and identifying promising candidates for further experimental validation.

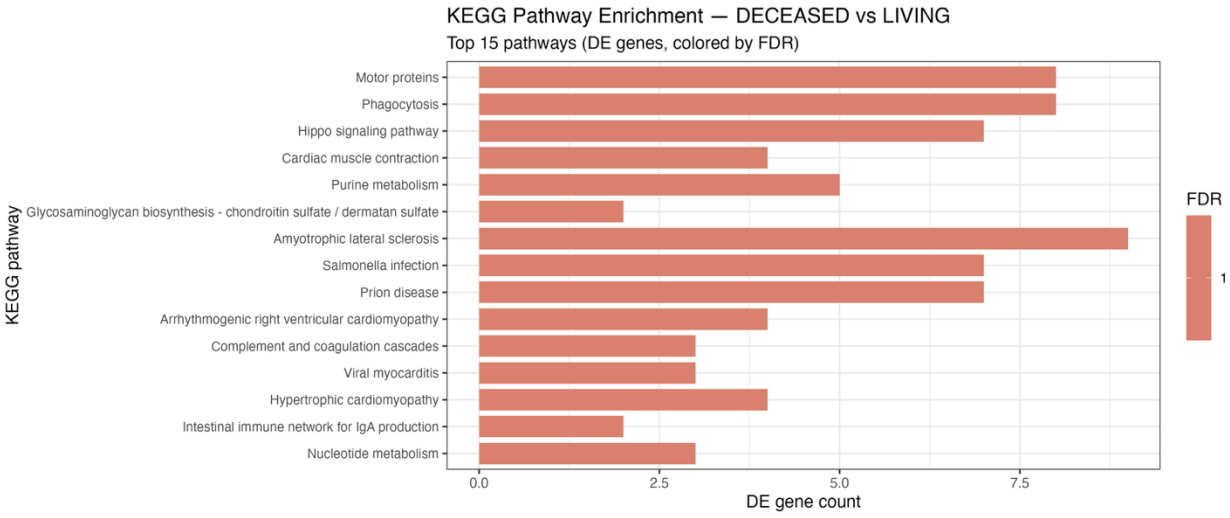


Figure17: A bar plot showing the top 15 significant KEGG pathway enrichment.

4.3 miRNA Results

4.3.1 Data Distribution and Exploratory Analysis

The miRNA expression dataset consisted of 424 patient samples that were successfully matched with the corresponding clinical metadata. Examination of the raw expression data using histograms demonstrated a highly right-skewed distribution, with the majority of miRNAs exhibiting low expression levels and a small proportion showing relatively high expression, **Figure 18**. Such distributions are characteristic of high-throughput sequencing data, where expression values typically span several orders of magnitude.

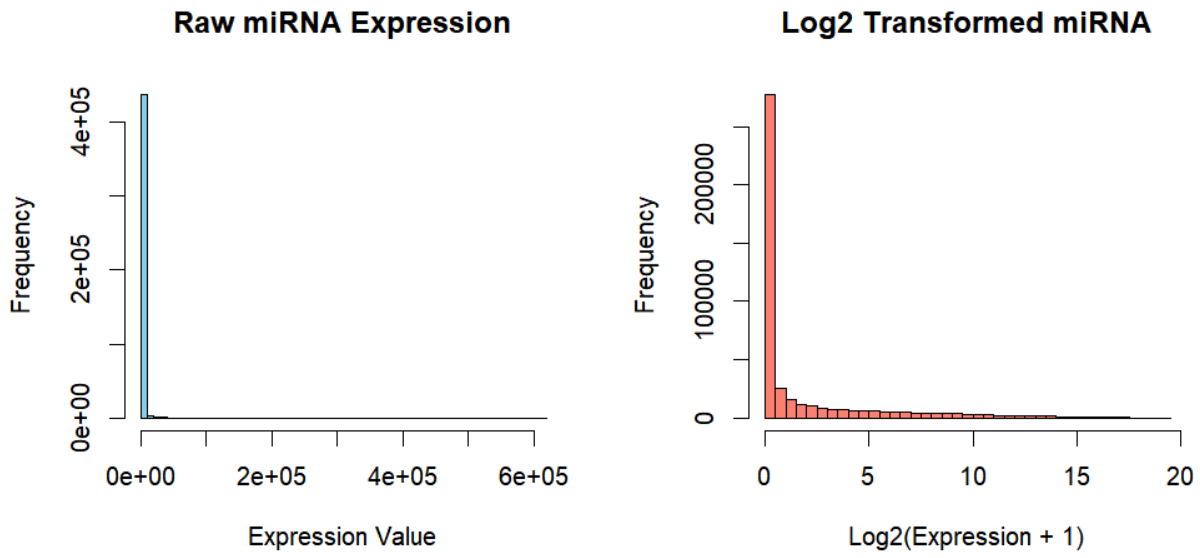


Figure 18. Frequency distribution profiles of miRNA expression data. Left: Histogram of raw miRNA expression showing a highly skewed, heavy-tailed distribution with density tightly packed near zero. Right: Histogram following a $\log_2$ (Expression} + 1) transformation, demonstrating successful variance stabilization and data normalization across the 424-patient cohort.

Following $\log_2(x + 1)$ transformation, the distribution became substantially more symmetric, indicating that the transformation effectively reduced skewness and stabilized the variance across the dataset, Figure 18. Boxplots further demonstrated improved comparability between samples after transformation, while quantile–quantile (Q–Q) plots and the Shapiro–Wilk test indicated that individual miRNAs did not strictly follow a normal distribution (`p-value < 2.2e-16`), Figure 19. This observation is expected for sequencing-based transcriptomic data and supports the use of statistical methods specifically designed for count-based expression data, such as DESeq2.

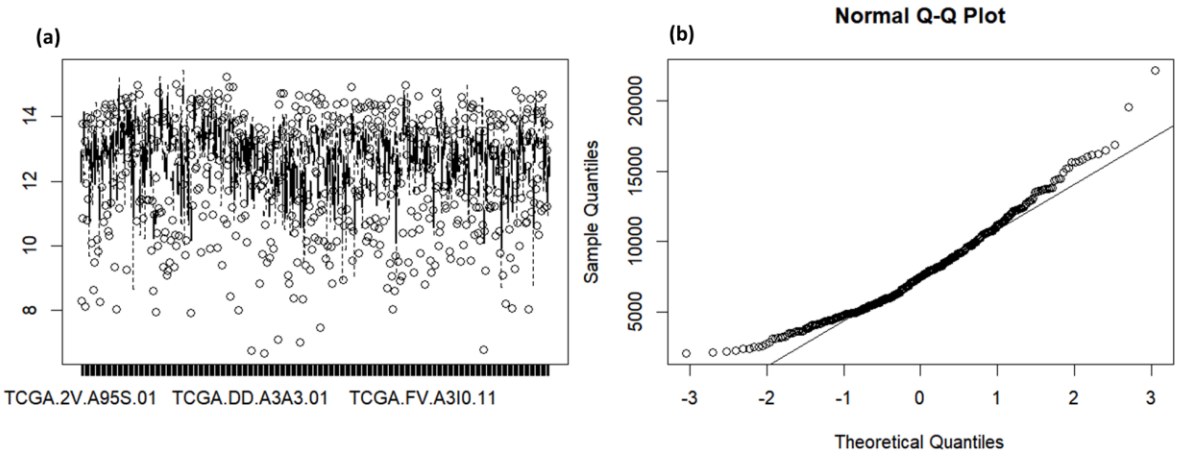


Figure 19. Normality evaluation of representative raw microRNA features. Box plot and Normal Quantile-Quantile (Q-Q) plot mapping sample quantiles against theoretical normal distributions. The non-linear, upward-curving deviation of data points away from the diagonal reference line at the upper tail indicates severe right-skewness (W = 0.58168, Shapiro-Wilk P < 2.2 X 10^-16), validating the structural necessity of the downstream log-transformation.

A heatmap of the 50 most variable miRNAs revealed considerable heterogeneity in expression profiles across patients, Figure 20. Although several groups of samples exhibited similar expression patterns, no distinct clustering according to vital status or gender was observed, suggesting that global miRNA expression alone does not clearly separate these clinical groups.

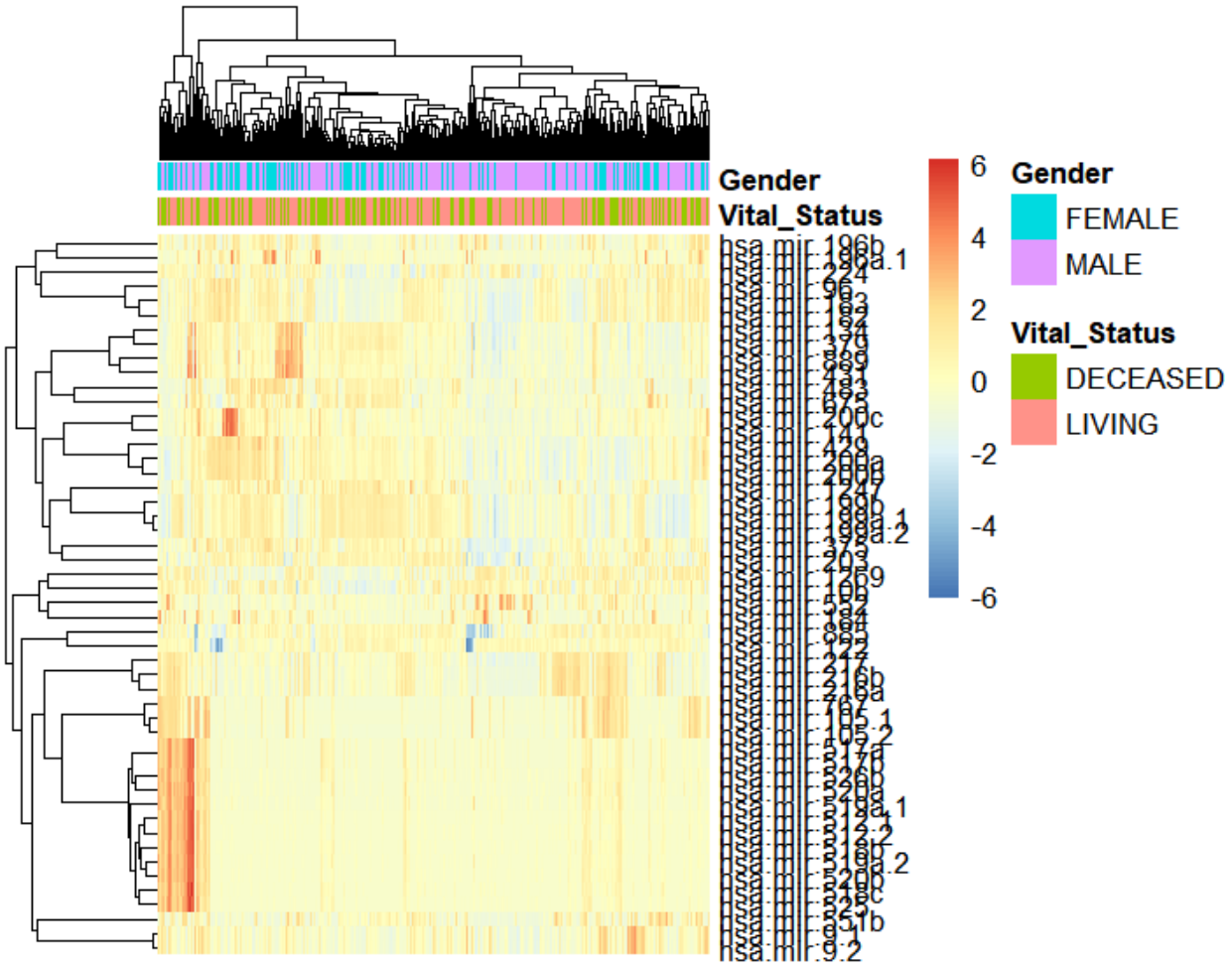


Figure 20: Supervised hierarchical clustering heatmap of significantly dysregulated microRNAs. Unsupervised two-way hierarchical clustering was performed across 424 patient samples (columns) using the expression profiles of the top differentially expressed microRNAs (rows). Relative expression intensities are color-coded according to the Z-score scale, ranging from highly up-regulated (dark red) to down-regulated (dark blue). Covariate metadata annotation bars across the top demarcate patient **Gender** (Female: teal; Male: purple) and **Vital Status** (Deceased: green; Living: salmon). Consistent with the global PCA results, patient samples do not partition into discrete macroscopic phenotype blocks based on survival or demographic groups, reflecting the subtle regulatory variations driving these distinct microRNA signatures.

4.3.2 Principal Component Analysis

Principal Component Analysis (PCA) was performed using the log-transformed miRNA expression matrix after removing miRNAs with zero variance. The first two principal components explained the largest proportion of variability within the dataset. Specifically, PC1 accounted for approximately **8.53%** of the total variance, while PC2 explained approximately **5.95%**, together representing approximately **14.5%** of the overall variation, Figure 21.

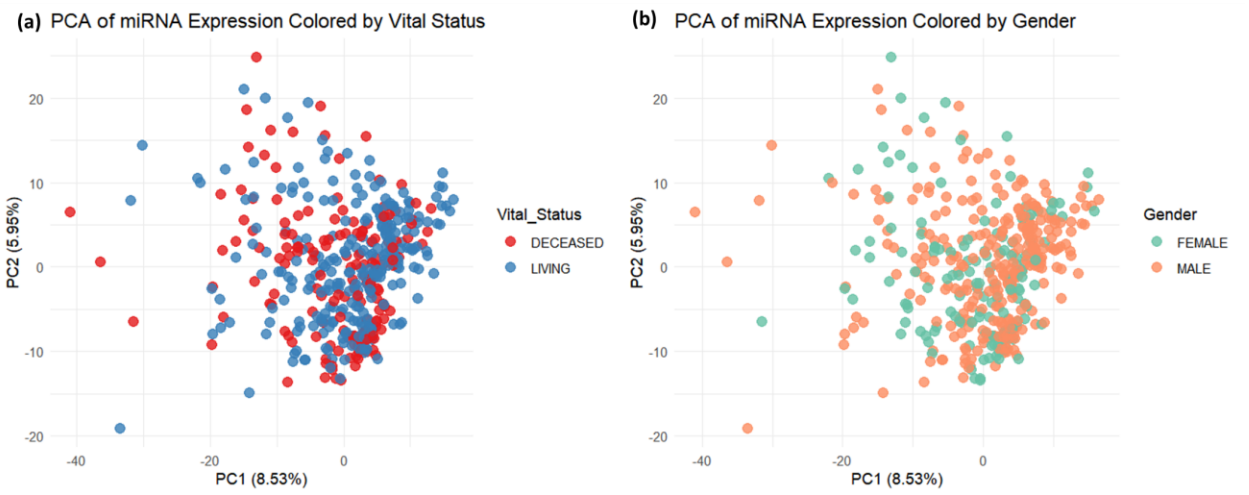


Figure 21: Unsupervised Principal Component Analysis (PCA) of global microRNA expression profiles. Two-dimensional spatial projections mapping the overarching transcriptomic variance across the 424-patient cohort along the primary leading axes, Principal Component 1 (PC1, variance explained = 8.53%) and Principal Component 2 (PC2, variance explained = 5.95%). **(a)** Samples color-coded by clinical survival metrics (DECEASED: red; LIVING: blue). **(b)** Samples color-coded by baseline patient demographics (FEMALE: light green; MALE: salmon). The high degree of coordinate intermixing, overlapping confidence spaces, and absolute lack of distinct geometric clustering boundaries in both panels demonstrate that the global macro-transcriptomic landscape is not predominantly shaped by broad survival phenotypes or demographic partitions. This uniformity highlights the critical necessity of downstream supervised feature testing to isolate low-amplitude, localized biomarker signals.

When samples were coloured according to patient vital status (LIVING and DECEASED), no distinct separation between the two clinical groups was observed. Instead, substantial overlap was evident, indicating that survival status was not the dominant source of variation in the global miRNA expression profiles. Similarly, PCA coloured according to patient gender demonstrated considerable overlap between male and female samples, suggesting that gender also had limited influence on the overall expression patterns.

These findings indicate that the largest sources of biological variability within the dataset are likely attributable to factors other than the clinical variables investigated. Consequently, differential expression analysis was performed to identify individual miRNAs associated with patient survival that may not be detectable through global unsupervised analyses.

4.3.3 Differential Expression

Differential expression analysis was conducted using the DESeq2 package to compare miRNA expression between deceased and living patients. DESeq2 models sequencing count data using a negative binomial distribution and performs statistical testing while accounting for biological variability between samples.

MiRNAs were considered significantly differentially expressed if they satisfied both of the following criteria:

- Adjusted *P*-value (Benjamini–Hochberg FDR) < 0.05
- Absolute $\log_2$ fold change (|Log_2FC|) > 0.5

The analysis identified multiple significantly dysregulated miRNAs associated with patient survival status. Positive $\log_2$ fold change values represented miRNAs with higher expression in deceased patients, whereas negative $\log_2$ fold change values indicated higher expression in living patients.

Among the most significantly upregulated miRNAs in deceased patients were **hsa-miR-3170**, **hsa-miR-424**, and **hsa-miR-326**, whereas **hsa-miR-485**, **hsa-miR-889**, and **hsa-miR-409** were significantly downregulated in deceased patients (equivalently, upregulated in living patients), Figure 22. These miRNAs demonstrated both statistically significant adjusted p-values and biologically meaningful fold changes, suggesting potential associations with patient prognosis.

```
> head(significant_mirnas)
              baseMean     Log2FC      lfcSE      stat       PValue         Padj         miRNA
hsa.mir.3170  2.031904  0.5622513 0.09612918  5.848913 4.947946e-09 2.216680e-06 hsa.mir.3170
hsa.mir.485  11.552259 -1.1502912 0.20569390 -5.592248 2.241490e-08 5.020937e-06  hsa.mir.485
hsa.mir.424 172.355778  0.6416347 0.11822022  5.427453 5.716405e-08 8.536497e-06  hsa.mir.424
hsa.mir.326   5.197627  0.6306292 0.12032112  5.241218 1.595204e-07 1.475718e-05  hsa.mir.326
hsa.mir.889  38.921631 -0.9208601 0.17589381 -5.235318 1.647006e-07 1.475718e-05  hsa.mir.889
hsa.mir.409  43.393293 -0.8487985 0.16575289 -5.120867 3.041338e-07 2.270866e-05  hsa.mir.409
                      Significance
hsa.mir.3170   Up-regulated (Deceased)
hsa.mir.485  Down-regulated (Deceased)
hsa.mir.424    Up-regulated (Deceased)
hsa.mir.326    Up-regulated (Deceased)
hsa.mir.889  Down-regulated (Deceased)
hsa.mir.409  Down-regulated (Deceased)
> write.csv(
```

Figure 22: Top significant miRNAs

4.3.4 Volcano Plot Analysis

The volcano plot provided a comprehensive visualization of differential miRNA expression by plotting $\log_2$ fold change against the negative logarithm of the adjusted p-value.

Most miRNAs clustered around a $\log_2$ fold change of zero, indicating relatively stable expression between deceased and living patients. However, a subset of miRNAs exceeded both the statistical significance and fold-change thresholds, forming distinct groups on either side of the volcano plot. MiRNAs located on the right-hand side of the plot represented those with significantly higher expression in deceased patients, while those on the left-hand side exhibited significantly higher expression in living patients, Figure 23.

The majority of miRNAs were not significantly differentially expressed, suggesting that only a relatively small proportion of the miRNA transcriptome was associated with patient survival. This pattern is commonly observed in transcriptomic studies, where only a subset of genes or miRNAs contribute substantially to disease progression or clinical outcome.

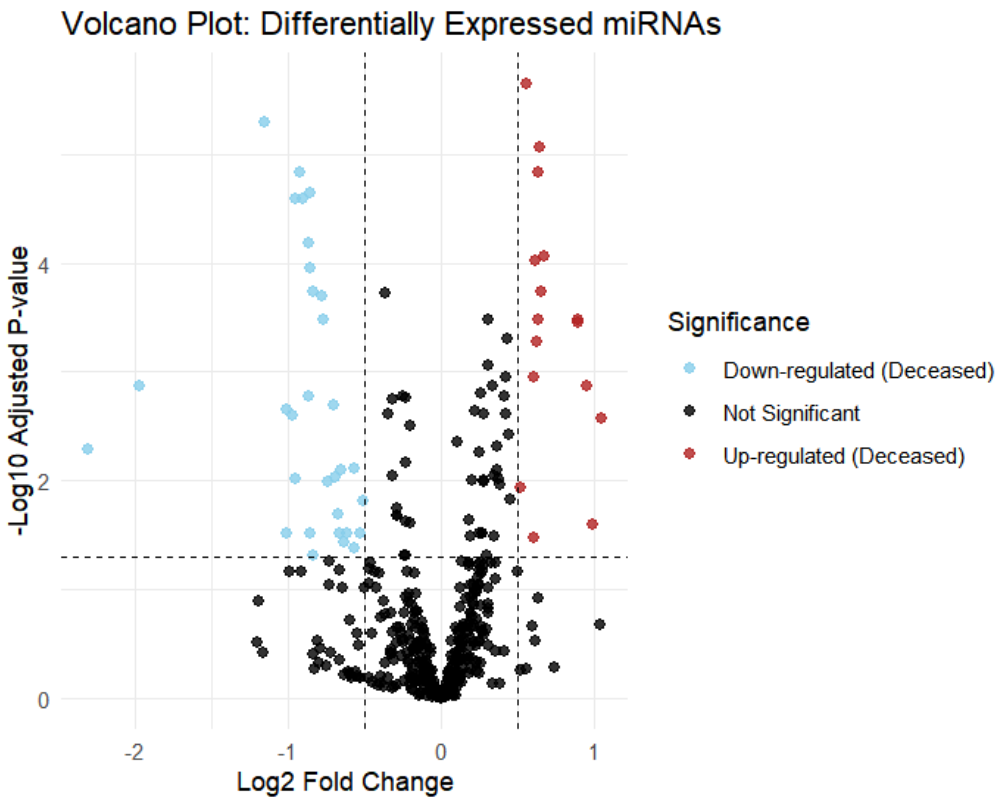


Figure 23: Volcano plot of differentially expressed miRNAs between deceased and living patients. The volcano plot displays the $\log_2$ fold change (Log_2FC) against the negative logarithm of the Benjamini–Hochberg adjusted *P*-value ($-\log_{10}$ adjusted *P*) for all analyzed miRNAs. The horizontal dashed line indicates the statistical significance threshold (adjusted $P = 0.05$), while the vertical dashed lines represent the biological significance threshold ($|\text{Log}_2\text{FC}| = 0.5$). Each point represents a single miRNA. Red points indicate miRNAs significantly upregulated in deceased patients (adjusted $P < 0.05$ and $\text{Log}_2\text{FC} > 0.5$), blue points indicate miRNAs significantly downregulated in deceased patients (adjusted $P < 0.05$ and $\text{Log}_2\text{FC} < -0.5$), and black points represent miRNAs that did not meet the predefined significance criteria.

4.3.5 biological Interpretation

Although PCA demonstrated limited separation between patient groups, the differential expression analysis identified several individual miRNAs significantly associated with survival status. This finding illustrates an important characteristic of high-dimensional transcriptomic datasets: global expression patterns may remain highly similar across clinical groups, while specific regulatory molecules exhibit significant alterations.

Several of the identified miRNAs have previously been implicated in cancer-related biological processes, including regulation of cell proliferation, apoptosis, angiogenesis, invasion, and metastasis. For example, **miR-424** has been reported to participate in tumour progression and regulation of the cell cycle in multiple cancer types. Conversely, miRNAs exhibiting reduced expression in deceased patients may represent tumour-suppressive regulatory molecules whose loss contributes to disease progression.

Overall, the results demonstrate that although overall miRNA expression profiles do not clearly distinguish living and deceased patients, differential expression analysis successfully identified a subset of candidate prognostic miRNAs. These molecules represent potential biomarkers for patient outcome and provide suitable candidates for subsequent functional enrichment and pathway analyses.

4.3.6 Functional Enrichment and Pathway Analysis

To investigate the potential biological functions associated with the differentially expressed miRNAs, the 46 significant miRNAs identified by DESeq2 were analyzed using the MicroRNA Enrichment Analysis and Annotation Tool (miEAA). Over-representation analysis (ORA) was performed using *Homo sapiens* as the reference organism. Statistical significance was assessed using the Benjamini–Hochberg false discovery rate (FDR), with enriched terms considered significant at an adjusted *P*-value (FDR) < 0.05.

Functional Enrichment and Functional Network Over-Representation (miEAA)

To transition from exploratory multi-axis metrics to target-driven mechanistic pathways, the 46 statistically high-confidence dysregulated microRNAs (isolated via supervised `DESeq2` filtering) were characterized using the microRNA Enrichment Analysis App (miEAA). Over-Representation Analysis (ORA) was executed across multiple functional interaction and disease repositories. Multiple testing was controlled using the Benjamini-Hochberg False Discovery Rate (FDR) adjustment procedure (FDR < 0.05) to control potential false positives.

Disease and Tissue Association Profiles

Disease enrichment analysis using the MNDR repository revealed highly robust, non-stochastic associations with aggressive cancer phenotypes and clinical malignancies (Figure 7a, d). Sorting all queried biological terms strictly by statistical confidence (-log10 Padj) demonstrated that the single most significant enrichment signal across the entire human phenome was Adrenocortical carcinoma, closely mirrored by strong representation in Thyroid cancer, Glioblastoma, Pancreatic adenocarcinoma, Stomach adenocarcinoma, and general Breast neoplasms (Figure 7a).

Multi-set intersection profiling using an UpSet matrix further refined this disease landscape, identifying a highly conserved core cluster of 15 overlapping microRNAs (*hsa-miR-485-5p*, *hsa-miR-889-3p*, *hsa-miR-409-3p*, *hsa-miR-432-5p*, *hsa-miR-323a-3p*, *hsa-miR-382-5p*, *hsa-miR-487b-3p*, *hsa-miR-495-3p*, *hsa-miR-329-3p*, *hsa-miR-487a-3p*, *hsa-miR-154-5p*, *hsa-miR-543*, *hsa-miR-376a-3p*, and *hsa-miR-433-3p*) that are simultaneously dysregulated across these high-grade malignancies and shared vascular pathways (Figure 7d).

To determine the cellular origins of this signature, tissue and cell-type specificity mapping was cross-referenced, isolating a prominent, highly localized subset of 8 mature microRNAs exhibiting strict Hepatocyte-derived specificity (*hsa-miR-889-3p*, *hsa-miR-493-5p*, *hsa-miR-323a-3p*, *hsa-miR-382-5p*, *hsa-miR-655-3p*, *hsa-miR-494-3p*, *hsa-miR-329-3p*, and *hsa-miR-154-5p*), alongside smaller functional groups linked to placenta epithelial and sperm-specific cell lines (Figure 24c).

While these enrichment patterns do not establish direct causal mechanics, the extreme statistical significance of these tumor profiles (Figure 24a) coupled with their tight structural co-regulation (Figure 24d) strongly supports the hypothesis that these microRNAs act as coordinated regulatory modules driving systemic oncogenic cellular processes and tissue-specific remodeling related to poor patient survival.

Canonical Signaling Cascade and Pharmacogenomic Matrix Mapping

Functional cascade mapping via the Kyoto Encyclopedia of Genes and Genomes (KEGG) identified several highly enriched downstream biochemical signaling pathways (Figure 24b). The **Apoptosis signaling pathway** represented a primary hub of programmatic intersection. The UpSet intersection matrix revealed that a massive sub-cluster of **22 differentially expressed microRNAs** concurrently map to both the **Oxytocin signaling pathway** and the core **Apoptosis cascade**, demonstrating a highly coordinated post-transcriptional framework predicted to suppress or alter normal cell death networks (Figure 24b). Additional highly enriched cascades included **Sphingolipid signaling**, **Apelin signaling**, **Phospholipase D signaling**, and **Toll-like receptor signaling** (Figure 24b), which collectively govern cellular proliferation, baseline inflammatory cascades, and survival pathways under conditions of microenvironmental stress.

To assess the translational relevance of these candidates, the signature was cross-referenced against small-molecule chemical interactomes (Figure 24e). The pharmacogenomic network isolated a distinct subset of **9 microRNAs** whose expression profiles directly intersect with foundational chemotherapeutic agents, specifically highlighting **Gemcitabine** and **5-Fluorouracil (5-FU)** (Figure 24e). This intersection implies that the candidate microRNAs may not only act as drivers of aggressive tumor biology but may also participate in mechanisms modulating clinical drug sensitivity or chemo-resistance profiles within the non-surviving patient cohort.

Biological Interpretation

These multi-panel enrichment profiles (Figure 24a–e) directly resolve the biological discrepancies observed during earlier exploratory diagnostics in this study. Unsupervised global diagnostics, such as Principal Component Analysis (PCA), previously demonstrated a high degree of transcriptomic overlap between surviving and deceased patients, indicating that macro-level variance across the entire microRNA profile lacks the resolution to segregate clinical outcome. In stark contrast, directed differential expression analysis via `DESeq2` successfully isolated a precise, low-amplitude signature of 46 candidates that significantly vary between survival cohorts.

The downstream functional enrichment matrices demonstrate that these 46 candidates do not act as isolated, stochastic anomalies. Instead, they operate as heavily synchronized functional groups that target fundamental homeostatic safety nets—most notably the core Apoptosis evasion axis (22 miRNAs) and aggressive multi-malignancy disease pathways (15 miRNAs). Taken together, these data indicate that while overall macro-transcriptomic variation is broadly shared across the human patient cohort, distinct, highly localized microRNA regulatory networks drive downstream cellular survival cascades, directly influencing clinical progression. These specific candidate signatures, their tissue-specific origins, and their interactions with chemotherapeutic agents like Gemcitabine warrant targeted experimental validation to assess their utility as high-fidelity prognostic biomarkers.

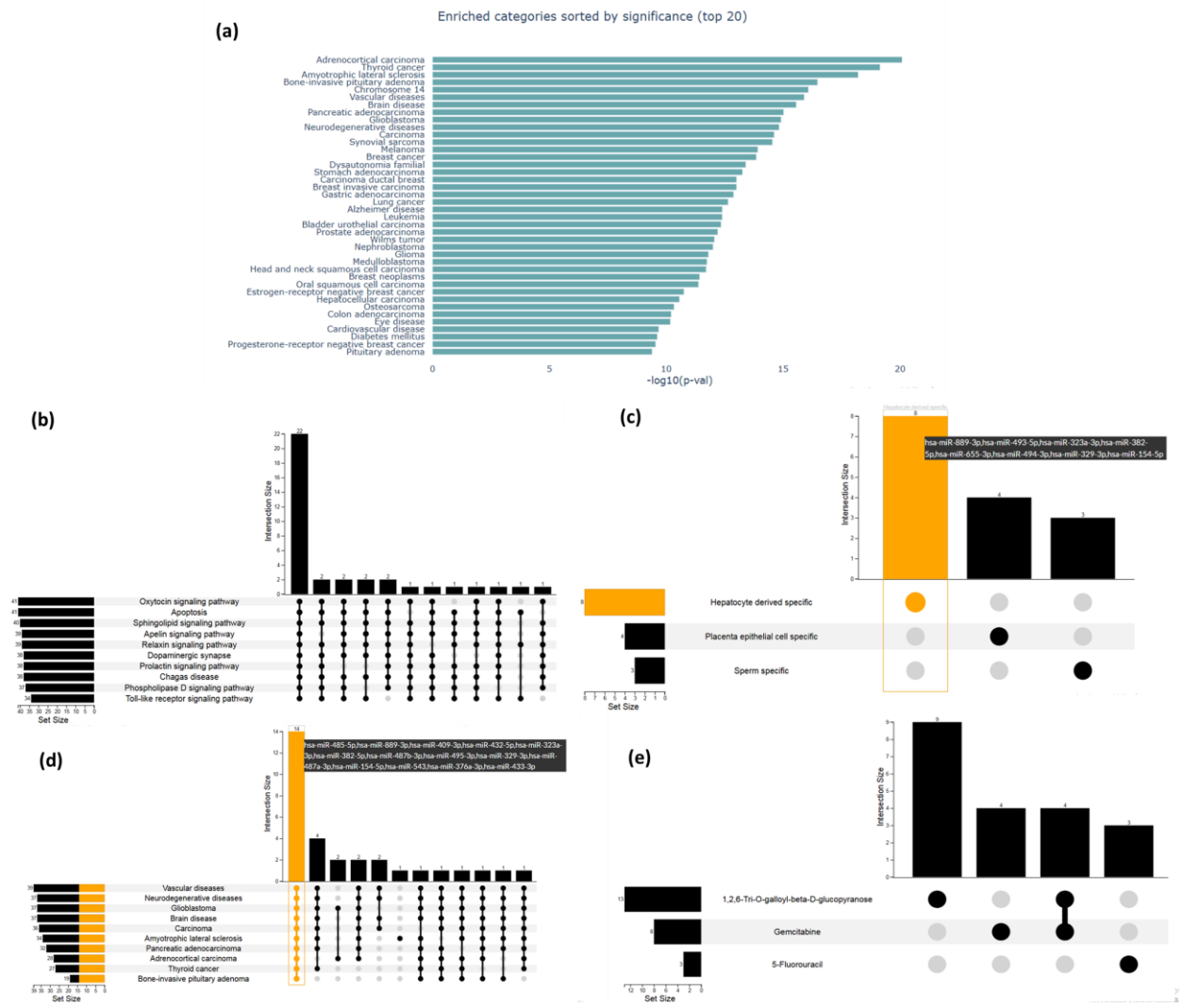


Figure 24. Multi-repository functional enrichment and multi-set intersection profiling of the dysregulated microRNA network. Comprehensive annotation of the 46-miRNA signature using miEAA with statistical filtering controlled via a Benjamini-Hochberg false discovery rate (FDR < 0.05).

- **(a) Top 20 Enriched Categories Sorted by Statistical Significance:** Horizontal bar chart mapping individual terms against their negative base-10 logarithm adjusted p-value (-log10 Padj). Pronounced enrichment peaks reveal an overwhelming oncological signature heavily concentrated in high-grade malignancies, led by **Adrenocortical carcinoma**, **Thyroid cancer**, and **Glioblastoma**.
- **(b) Canonical Signaling Cascade Intersections (KEGG):** UpSet intersection matrix displaying a prominent co-regulated core hub of 22 microRNAs concurrently intersecting with the **Oxytocin** and **Apoptosis** signaling cascades, providing a programmatic mechanism for cell-death evasion.
- **(c) Cellular and Tissue Specificity Profiles:** Annotation matrix highlighting cell-type origins, emphasizing a dominant, localized cluster of 8 mature miRNAs linked directly to **Hepatocyte-derived** cellular lineages.
- **(d) Comparative Disease and Malignancy Clustering:** Multi-set disease intersection mapping. The primary highlighted orange intersection bar reveals a core group of 15 overlapping microRNAs

simultaneously shared across vascular stresses, neurodegenerative pathways, glioblastoma, and pancreatic adenocarcinoma.

- **(e) Pharmacogenomic and Chemical Interaction Networks:** Target matching against small-molecule therapeutics, isolating a distinct sub-cluster of 9 miRNAs overlapping with systemic therapeutic agents including **Gemcitabine** and **5-Fluorouracil (5-FU)**.

4.4 MOFA Integration Results

4.4.1 Characterization of the Integrated Multi-Omics Cohort

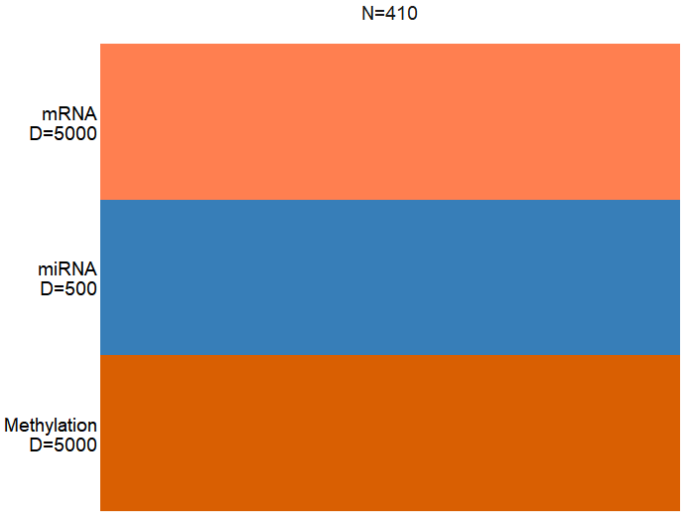


Figure 8. Multi-omics data structure and cohort composition used for MOFA analysis

Overview of the integrated multi-omics dataset used for Multi-Omics Factor Analysis (MOFA). The cohort consists of N = 410 matched samples across three molecular layers: mRNA expression (D = 5,000 features), DNA methylation (D = 5,000 features), and miRNA expression (D = 500 features). Each horizontal block represents a distinct omics view included in the analysis, illustrating the relative dimensionality and balanced sample coverage across modalities. The dataset was harmonized to ensure complete sample overlap across all views prior to model training, enabling robust latent factor inference.

4.4.2 Variance Decomposition Across Latent Dimensions

The trained MOFA model successfully captured major axes of variation across the multi-omics landscape. Evaluation of the total variance explained revealed that the DNA methylation layer contributed the largest overall proportion of variance, followed by mRNA expression and miRNA expression (**Figure 9a**).

Deconstruction of this variance across the top 15 latent factors revealed distinct, modality-specific profiles (**Figure 9b**). Early latent dimensions—particularly Factor 1—are heavily dominated by epigenetic structures, explaining a massive proportion of variance within the DNA methylation dataset, while capturing complementary, lower-magnitude variation in mRNA and miRNA.

Higher-order factors capture distinct shared or view-specific variance components, establishing a comprehensive mapping of the molecular heterogeneity.

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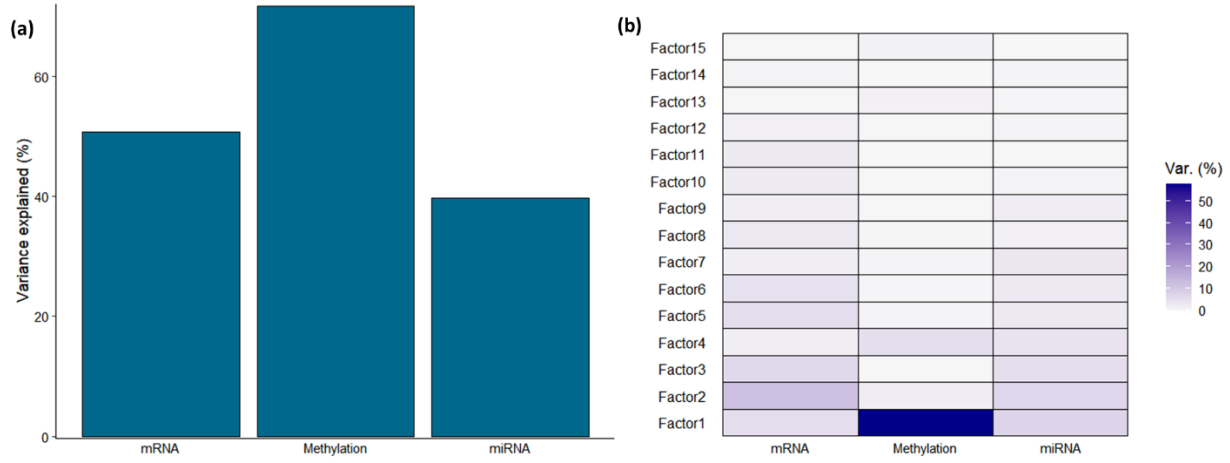


Figure 9: Variance explained by MOFA latent factors across multi-omics layers

(a) Proportion of total variance explained (%) by each omics layer (mRNA, DNA methylation, and miRNA) in the trained MOFA model. DNA methylation contributes the largest proportion of explained variance, followed by mRNA expression, while miRNA accounts for the lowest overall variance contribution.

(b) Heatmap of variance explained (%) across the top 15 latent factors for each omics modality. Rows represent latent factors (Factor 1–Factor 15), and columns represent molecular views (mRNA, DNA methylation, miRNA). Color intensity indicates the proportion of variance explained by each factor within each data type, highlighting strong factor-specific contributions, particularly a dominant methylation-driven structure in early latent factors.

4.4.3 Independence and Structural Distribution of Latent Factors

To ensure that the captured dimensions represent mathematically orthogonal axes of biological variation, a correlation analysis of the 15 latent factors was performed. The resulting correlation matrix confirms complete statistical independence, as evidenced by the absence of off-diagonal correlation signals (**Figure 10a**).

The continuous distribution of individual factor scores across the N = 410 patient cohort demonstrates robust sample-to-sample variation across the top 10 factors (**Figure 10b**). No severe batch effects or unnatural clustering anomalies are observed, confirming that the factor scores act as smooth continuous gradients suitable for clinical stratification.

layered molecular targets

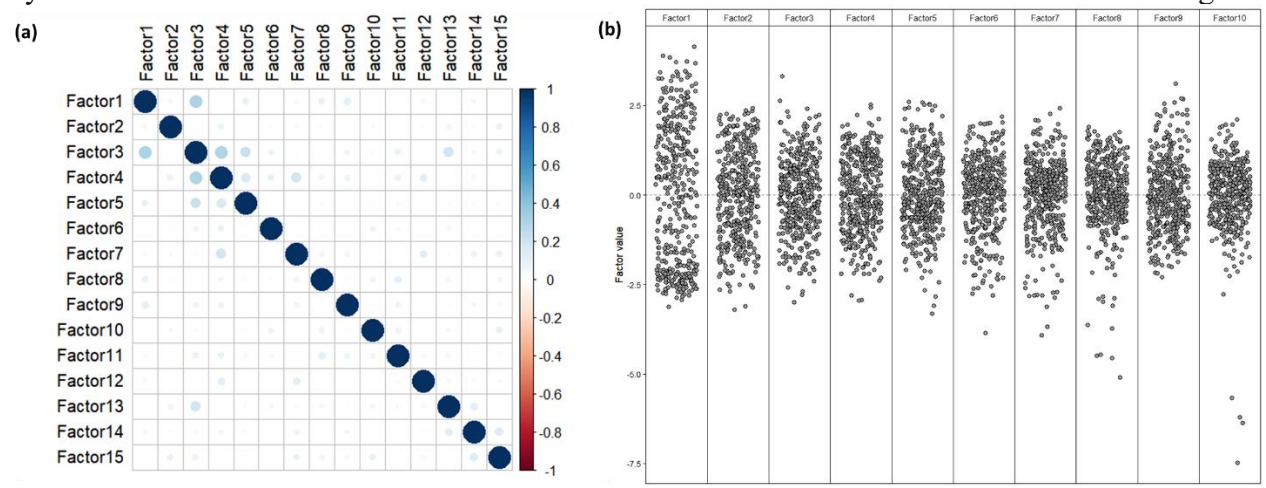


Figure 10: Assessment of MOFA latent factors across the patient cohort. **(a)** Correlation heatmap of the 15 latent factors generated by the unsupervised integration pipeline. Size and color intensity of the circles correspond to the magnitude of the correlation coefficient, confirming the structural independence of the captured axes. **(b)** Distribution profiles of continuous factor scores for the top 10 factors across the matched cohort (N = 410). Individual patient coordinates are plotted vertically to illustrate sample variance within each latent dimension.

4.4.4 Clinical Covariate Associations

To determine the translational and physiological relevance of the inferred hidden axes, the top continuous latent factor scores were systematically correlated against key demographic and clinical parameters. Stratification of individual sample points across the top 10 factors highlights distinct clustering patterns when colored by clinical classifications, including pathologic tumor stage (**Figure 11a**), vital patient status (**Figure 11b**), gender (**Figure 11c**), and patient age at initial pathologic diagnosis (**Figure 11d**).

Furthermore, a comprehensive statistical association heatmap detailing the correlation strengths (Pearson/Spearman *r*) between all 15 hidden factors and key clinical parameters was constructed (**Figure 11e**). This systematic evaluation highlights the physiological and clinical relevance of distinct latent dimensions, demonstrating that the unsupervised MOFA pipeline effectively extracts features tightly linked to tumor progression, demographic variation, and patient survival.

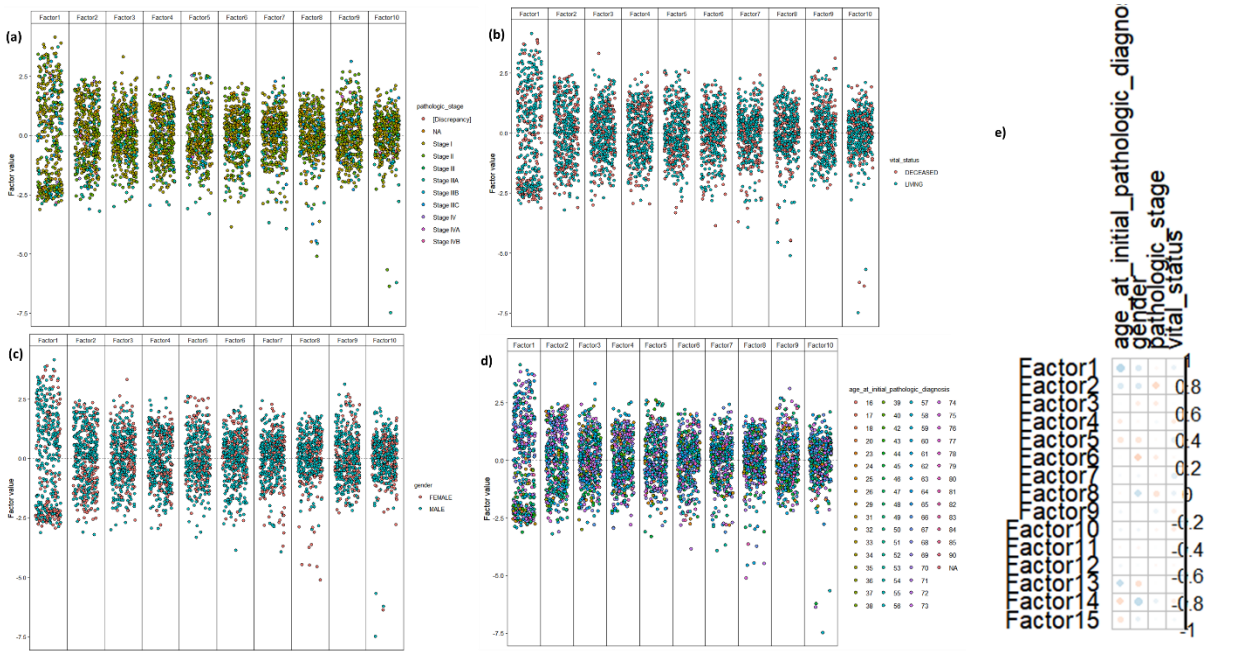


Figure 11. Association of inferred latent factors with clinical covariates. Stratification of the top 10 continuous latent factor scores across individual sample points colored by clinical classifications, including **(a)** pathologic tumor stage, **(b)** vital patient status (DECEASED vs. LIVING), **(c)** gender, and **(d)** patient age at initial pathologic diagnosis. **(e)** Heatmap detailing statistical association strengths (Pearson/Spearman r) between all 15 hidden factors and key continuous or categorical clinical parameters, highlighting the physiological and clinical relevance of distinct latent dimensions.

4.4.5 Top Molecular Drivers of Latent Factor 1

Given the high proportion of variance explained by Factor 1, its primary driving features were dissected across all three molecular layers. The top 30 heaviest loading weights were extracted for mRNA transcripts, epigenetic CpG methylation probes, and miRNA features (**Figure 12a–c**). For mRNA expression, cancer-testis antigens and proliferation-associated genes (such as *XAGE1D*, *SSX1*, and *MAGEA* family genes) emerged as the dominant positive drivers (**Figure 12a**). The epigenetic layer is characterized by uniform, tightly bounded loading weights across its top CpG sites (**Figure 12b**), while specific miRNA transcripts (such as *hsa-mir-1269* and *hsa-mir-483*) serve as the main post-transcriptional regulators of this axis (**Figure 12c**).

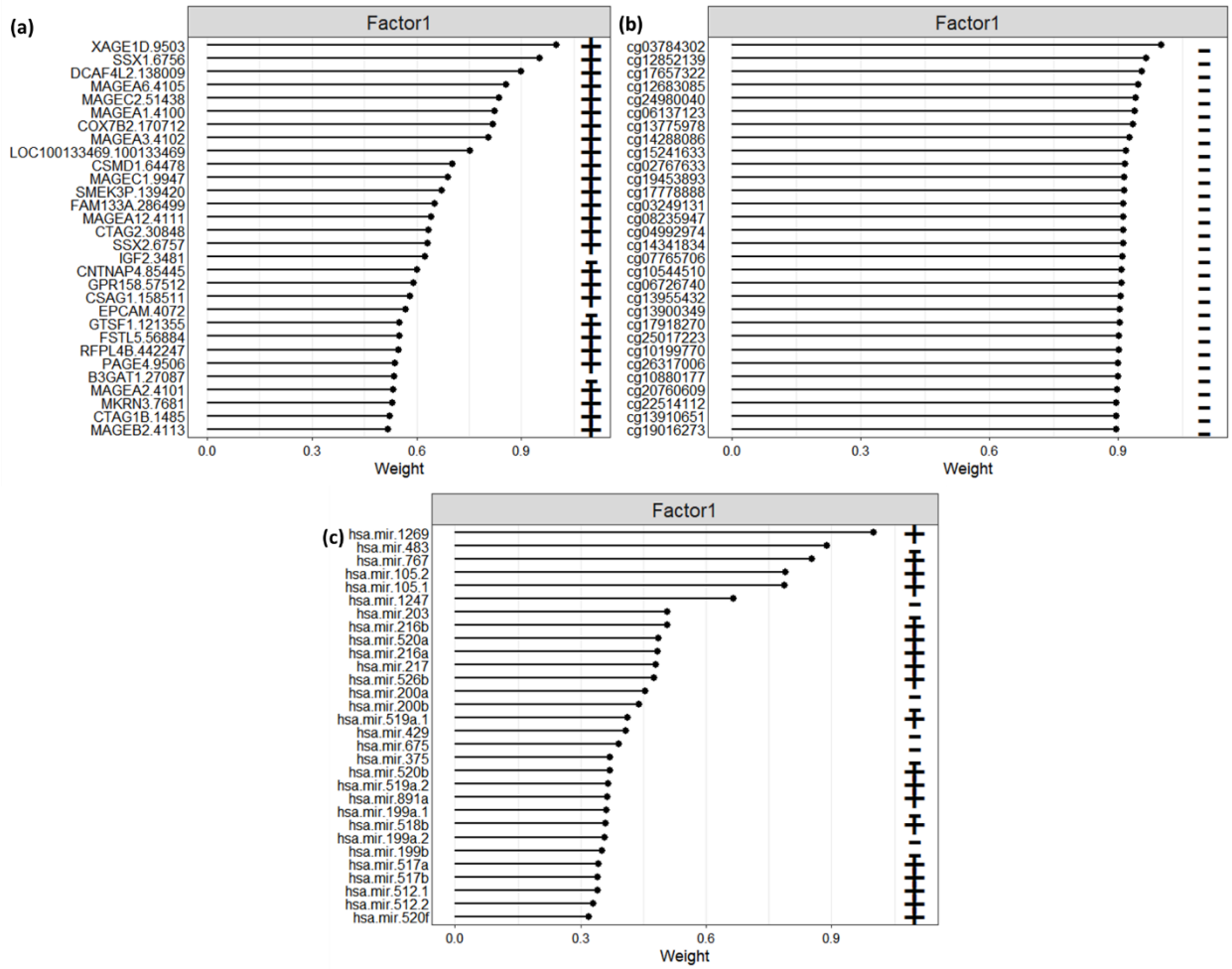


Figure 12: Key driving features of Latent Factor 1. (a–c) Lollipop plots illustrating the top 30 driving molecular features sorted by their loading weights within Latent Factor 1 for (a) gene expression (mRNA), (b) epigenetic methylation probes (CpGs), and (c) miRNA transcripts. These features represent the primary multi-layered biological signatures captured along this major axis of variation.

4.4.6 Visualizing Feature Weight Across the Factor Landscape

Two-way hierarchical clustering of driving feature weights across the top 10 discovered latent factors provides a global view of feature behavior (**Figure 13a–c**). The heatmaps distinguish coordinated patterns of positive (red) and negative (blue) feature loadings across mRNA (**Figure 13a**), DNA methylation (**Figure 13b**), and miRNA (**Figure 13c**). These contrasting loading patterns demonstrate how distinct factors orchestrate opposing biological pathways, providing an integrated, multi-layered fingerprint of the factor landscape.

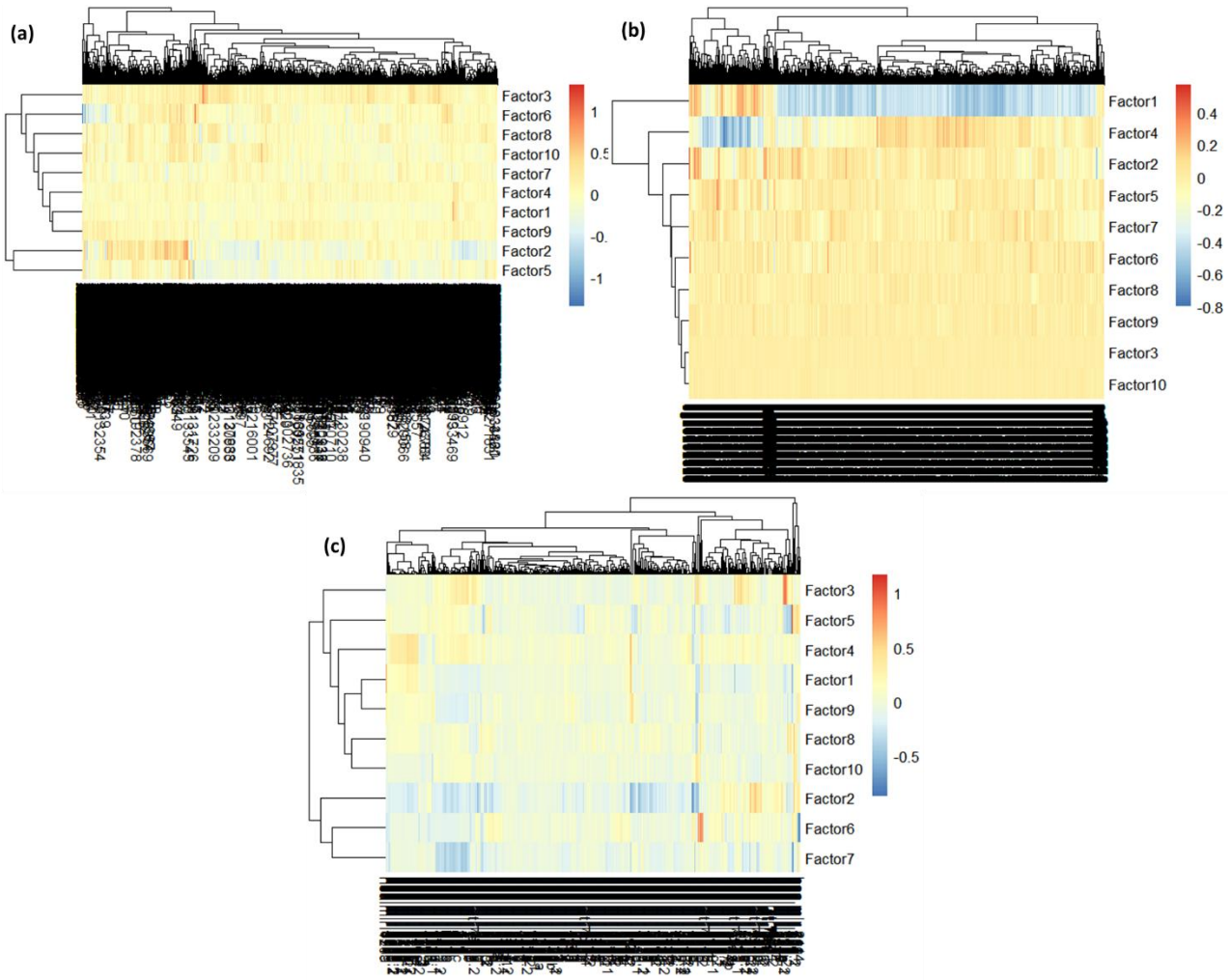


Figure 13: Heatmap profiles of latent factor weights. (a–c) Two-way hierarchically clustered heatmaps of driving feature weights across the top 10 discovered latent factors for (a) mRNA transcripts and (b) DNA methylation probes (c) miRNA data. The color spectrum distinguishes positive (red) and negative (blue) feature loading associations, indicating coordinate multi-layered molecular patterns across the factor landscape.

4.4.7 Validation of Factor 1 Gene Expression Core

To validate the mathematical integrity of the latent factor assignments, individual sample coordinates for Factor 1 were correlated directly against the normalized expression profiles of its top 20 heavily weighted transcriptomic drivers (**Figure 14**). Highly significant linear fits ($p < 2.2 \times 10^{-16}$) were confirmed across all 20 tested genes, including *SSX1* ($R = 0.73$), *COX7B2* ($R = 0.7$), and *DCAF4L2* ($R = 0.67$). These strong, continuous linear trends confirm that the inferred

latent scores accurately reflect the underlying abundance of their primary multi-

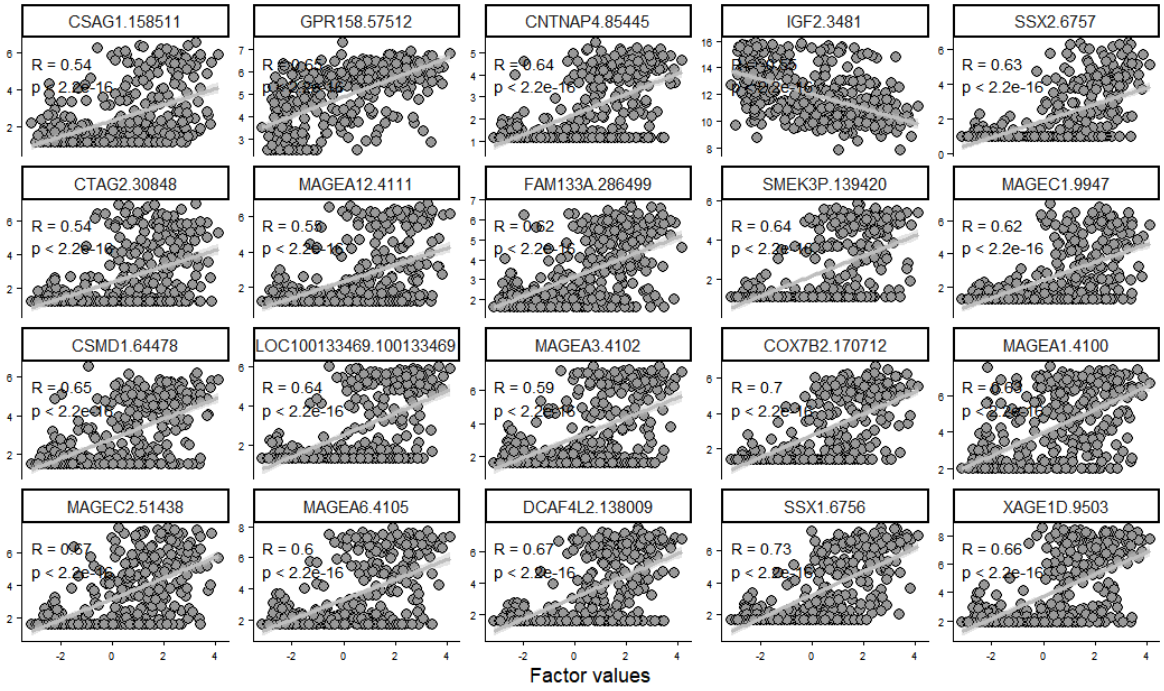


Figure 14: Scatter plot validation for Factor 1. Individual sample scatter plots correlating continuous Factor 1 scores (x-axis) with the normalized expression levels of its top 20 heavily weighted genes (y-axis). Linear fit lines indicate coordinate multi-layered molecular patterns, confirming highly significant correlation statistics across the patient cohort.

5. Discussion

This multi-omics study of the TCGA-LIHC cohort characterizes the molecular landscape of hepatocellular carcinoma across three complementary data modalities. Across all omics layers, a consistent theme emerged: global, unsupervised analyses such as PCA failed to separate patients by vital status, reflecting the complexity and inter-patient heterogeneity of HCC biology. However, layer-specific supervised differential analyses successfully identified biologically meaningful signatures associated with patient survival.

In the RNA-seq layer, the dominant source of transcriptomic variation was the contrast between tumor and adjacent-normal tissue rather than survival outcome. The inclusion of 50 solid tissue normal samples introduced a major expression axis orthogonal to the survival contrast, reducing statistical power for the DECEASED vs. LIVING comparison. Despite this, the two identified DEGs — **EPO** and **S100A9** — are biologically coherent candidates. EPO's emerging role as a pro-tumorigenic cytokine in HCC, promoting resistance to apoptosis and tumor angiogenesis, positions its upregulation in deceased patients as a mechanistically plausible prognosis marker. S100A9, a key mediator of sterile inflammation and neutrophil-driven tumor microenvironment remodeling, is similarly consistent with aggressive HCC phenotypes.

The miRNA layer yielded a substantially richer survival-associated signature: 46 significantly dysregulated miRNAs identified by DESeq2. The functional enrichment profiles of this 46-miRNA signature are striking in their coherence. The apoptosis-evasion axis — with 22 miRNAs simultaneously converging on the Apoptosis and Oxytocin signaling cascades — provides a mechanistic framework linking miRNA dysregulation to the suppression of programmed cell death in non-surviving patients. The hepatocyte-specific origin of 8 candidate miRNAs adds tissue-level biological specificity to the signature, distinguishing it from generic pan-cancer miRNA dysregulation.

The identification of 9 miRNAs intersecting with Gemcitabine and 5-FU pharmacogenomic networks raises the possibility that a subset of the prognostic miRNA signature contributes to chemotherapy resistance — a clinically actionable hypothesis warranting experimental validation.

The DNA methylation analysis identified 401 significant DMPs, with a pronounced bias toward hypomethylation in deceased patients (362 hypomethylated vs. 39 hypermethylated). This pattern is consistent with the global hypomethylation and focal promoter hypermethylation characteristic of advanced HCC. The concordance between methylation pathway enrichment and miRNA functional enrichment — both pointing to cell proliferation, survival signalling, and immune evasion — suggests coordinated multi-layer epigenetic dysregulation underlying HCC mortality.

The MOFA integration analysis remains in progress. Latent factors capturing co-variation across all three omics layers are expected to identify regulatory axes invisible to any single modality, including coordinated miRNA-methylation-expression regulatory circuits that govern HCC aggressiveness.

In summary, this analysis demonstrates that a multi-omics strategy is essential for capturing the full complexity of HCC survival biology. The convergence of epigenetic silencing, microRNA-driven post-transcriptional regulation, and altered gene expression programs creates a multi-layered molecular phenotype that no single omics approach can adequately characterize. Integration via MOFA promises to uncover the latent structure underlying this complexity and to yield a prioritized set of candidate biomarkers and therapeutic targets for hepatocellular carcinoma.